EINSTEIN METRICS WITH TORUS SYMMETRY ON S^4 AND $\mathbb{R}^4$

Abstract. This note gives direct proofs that, under an effective isometric T^2 action, Einstein metrics on smooth homotopy four-spheres are round and complete Ricci-flat metrics on smooth manifolds homeomorphic to $\mathbb{R}^4$ are Euclidean or positive one-center Taub–NUT, up to isometry, scaling, and orientation reversal.

1. Introduction

A Riemannian metric g is Einstein if $\mathrm{Ric}_g = \lambda g$ for some constant λ. The classification of such metrics is a basic problem in Riemannian geometry; see [2]. In dimension four, even the simplest underlying manifolds lead to natural rigidity questions. For S^4, the classical question is whether an Einstein metric must be round, up to isometry and scaling. For $\mathbb{R}^4$, the complete Ricci-flat examples include both the Euclidean metric and the Taub–NUT metric [13, 11]. In this note, we give proofs of the corresponding rigidity statements under an effective isometric T^2 action.

Theorem 1.1. *Let M be a smooth closed homotopy four-sphere, and let g be an Einstein metric on M. If (M, g) admits an effective isometric T^2 action, then (M, g) is isometric, after constant rescaling, to the unit round four-sphere.*

Liu proved the sphere rigidity statement under the additional assumption of nonnegative sectional curvature, and established the polar reduction for closed simply connected toric Einstein four-manifolds without that curvature assumption [17, Theorems 1.1–1.2]. The sphere case is also contained in Li–Sun's uniqueness theorem [14, Theorem 1.5]; they record a proof by Broder and Lafuente in [14, Remark D.19].

For complete Ricci-flat metrics on manifolds homeomorphic to $\mathbb{R}^4$, Chen–Li [5, Theorem 1.6] obtained the Euclidean or Taub–NUT classification under the curvature bound

$$|\mathrm{Rm}_g|(x) \le \frac{k(r(x))}{r(x)^2}, \qquad \int_1^\infty \frac{k(s)}{s}\, ds < \infty,$$

where r is distance from a fixed point and k is positive and nonincreasing, provided that the tangent cone at infinity is not $\mathbb{R} \times [0, \infty)$. Their theorem does not assume symmetry. Here we assume torus symmetry and impose no curvature-decay or asymptotic hypothesis.

Theorem 1.2. *Let (M^4, g) be a smooth complete Ricci-flat manifold homeomorphic to $\mathbb{R}^4$. If (M, g) admits an effective isometric T^2 action, then g is Euclidean or positive one-center Taub–NUT, up to isometry, constant rescaling, and orientation reversal.*

The noncompact conclusion can also be recovered from Li–Sun's end analysis and harmonic-map theory [14, 15], together with the energy and curvature estimates of

Date: October 8, 2026.

Weber [20] and Cheeger–Tian [4]. Our approach gives an intrinsic proof through Green functions, sharp harmonic-map estimates, and curvature transgression.

We briefly describe the two arguments. The torus-action classification of Orlik–Raymond [18] determines the orbit space and the collapsing circle along each boundary arc. In the compact case, the Einstein constant is positive, so we normalize $\mathrm{Ric}_g = 3g$. Write h for the orbit metric and $\rho = \sqrt{\det G}$ for the orbit-area function, where G is the Gram matrix of the two Killing fields. The reduced Einstein equation gives $K_h \le 1$ and

$$\nabla_h^2(-\log\rho) \ge 2h.$$

Thus $-\log\rho$ is a strictly convex exhaustion of the interior of the orbit space. We use this to show that its metric completion is CAT(1). The two boundary arcs join the same fixed points and are local geodesics. The CAT(1) property then forces the diameter to be at least π. Bonnet–Myers and Cheng's maximal-diameter theorem [6] give roundness.

In the Ricci-flat case, ρ is harmonic. Let τ be its harmonic conjugate, normalized to vanish at the fixed point p. We first identify the minimal positive Green function, normalized by $-\Delta_g G_p = 4\pi^2\delta_p$, as

$$G_p = \frac{1}{2\sqrt{\rho^2+\tau^2}}.$$

Li–Yau's Green-function estimate [16] then gives proper global Weyl coordinates, and Colding's gradient estimate [9] gives a sharp lower bound for the orbit metric. The normalized matrix $H = \rho^{-1}G$ defines a harmonic map into $\mathbb{H}^2$. A sharp asymptotic estimate for its τ-derivative separates the Euclidean case from a collapsing case. In the latter case, estimates for suitable torus Killing fields control the boundary terms in the Euler and Pontryagin transgression formulas. The Euler identity first gives finite curvature energy. The Pontryagin identity then forces one Weyl half to vanish, and simple connectivity gives a hyperkähler structure. A circle subgroup of the torus preserves the hyperkähler triple and is tri-Hamiltonian. Swann's complete classification [19, Theorem 3.1], with the center description in [10, Section 2.2], then applies. The topology of $\mathbb{R}^4$ forces a single center, leaving precisely the Euclidean and Taub–NUT cases.

Section 2 proves Theorem 1.1. Section 3 develops the global coordinates, analytic estimates, and curvature identities used to prove Theorem 1.2.

AI disclosure. This note arose from an experiment in mathematical research through interaction with ChatGPT. The author posed the problems and follow-up questions, while ChatGPT developed the proposed mathematical arguments and generated the initial manuscript. Its role was therefore substantial and was not limited to language editing. The author worked through the arguments and rewrote the main body in their own words, based on their understanding.

2. Einstein metrics on S^4

This section proves Theorem 1.1.

2.1. The action and orthogonal transitivity.

Write $\mathrm{Ric}_g = \lambda g$. The Einstein constant is positive. Indeed, a nonzero Killing field X generated by a circle subgroup

satisfies $\frac{1}{2}\Delta_g|X|_g^2 = |\nabla^g X|_g^2 - \mathrm{Ric}_g(X,X)$. Integrating gives

$$\int_M |\nabla^g X|_g^2 \, dV_g = \lambda \int_M |X|_g^2 \, dV_g,$$

which excludes $\lambda < 0$, and the equality would make a nonzero parallel Killing field, contradicting $H^1(M;\mathbb{R}) = 0$.

After rescaling, we can assume that $\mathrm{Ric}_g = 3g$. Orlik–Raymond applies because $H_1(M;\mathbb{Z}) = 0$: the quotient is a disk with no exceptional finite-isotropy orbits, and $\chi(M) = 2$ gives two fixed points. Their smooth equivariant classification identifies the smoothing and action with the standard sphere [18, §§4.2, 5.1–5.4]. Thus, up to a diffeomorphism and a torus automorphism, we can assume that the effective T^2-action is given by the standard action:

$$\left(e^{is_1}, e^{is_2}\right) \cdot (z_1, z_2, t) = \left(e^{is_1} z_1, e^{is_2} z_2, t\right),$$

where $s_1, s_2 \in \mathbb{R}/2\pi\mathbb{Z}$ are the torus action parameters and $M = S^4 = \{(z_1, z_2, t) \in \mathbb{C}^2 \times \mathbb{R} : |z_1|^2 + |z_2|^2 + t^2 = 1\}$.

Let $B = M/T^2$ be the orbit space and $\pi : M \to B$ be the quotient map. Then B is a closed 2-disk, the action is free over the interior B° and the boundary ∂B consists of two arcs Γ_1, Γ_2 joining two corners $P^\pm$. More precisely, one can identify Γ_1 with the orbit of the invariant 2-sphere $\{(z_1, 0, t) : |z_1|^2 + t^2 = 1\}$, Γ_2 with the orbit of the invariant 2-sphere $\{(0, z_2, t) : |z_2|^2 + t^2 = 1\}$, and $P^\pm$ with the orbit of the fixed points $\{(0, 0, \pm 1)\}$.

Let X, Y be the Killing fields of the two coordinate circles, and let $\alpha = g(X, \cdot)$ and $\beta = g(Y, \cdot)$ be their metric dual 1-forms. In these coordinates,

$$X(z_1, z_2, t) = (iz_1, 0, 0), \quad Y(z_1, z_2, t) = (0, iz_2, 0).$$

The twist 1-form associated with X is

$$\omega_X = *(\alpha \wedge d\alpha),$$

where $* = *_g$ is the Hodge star operator. Similarly, we have twist form ω_Y.

For a Killing field, the twist identity says

$$d\omega_X = 2 * (\alpha \wedge \mathrm{Ric}_g(X, \cdot)).$$

Remark 2.1. This twist identity also follows from the Hessian identity for a Killing field X:

$$\nabla^2_{V,W} X := \nabla_V \nabla_W X - \nabla_{\nabla_V W} X = R(V, X)W.$$

Taking the trace of this Hessian identity gives

$$\Delta X = -\mathrm{Ric}^\sharp(X).$$

The Killing equation gives $(d\alpha)_{ij} = 2\nabla_i \alpha_j$, and thus

$$(\delta d\alpha)_j = -\nabla^i (d\alpha)_{ij} = 2\mathrm{Ric}_{jk} X^k,$$

where $\delta = - * d *$ is the codifferential. Applying Hodge star gives

$$d(*d\alpha) = 2 * \mathrm{Ric}_g(X, \cdot).$$

Finally, use the facts that $\omega_X = \iota_X(*d\alpha)$, $\mathcal{L}_X(*d\alpha) = 0$ and $\mathcal{L}_X = d\iota_X + \iota_X d$.

By the Einstein assumption, ω_X, ω_Y are closed, i.e.

$$d\omega_X = 0, \quad d\omega_Y = 0.$$

Since T^2 is abelian, X, Y commute, and consequently $\mathcal{L}_Y\omega_X = \mathcal{L}_X\omega_Y = 0$. By Cartan's formula, $d(\iota_Y\omega_X) = d(\iota_X\omega_Y) = 0$, thus the scalars $\iota_Y\omega_X$ and $\iota_X\omega_Y$ are constants. Since X and Y vanish at some point, we know $\iota_Y\omega_X = \iota_X\omega_Y \equiv 0$. Since $\iota_X(*\eta) = *(\eta \wedge \alpha)$ for a differential form η, we know

$$\alpha \wedge \beta \wedge d\alpha = 0, \quad \alpha \wedge \beta \wedge d\beta = 0.$$

Consider the distribution $\mathcal{H} = \ker\alpha \cap \ker\beta = (\mathrm{span}\{X, Y\})^\perp$ on the principal region $M^\circ := \pi^{-1}(B^\circ)$. Then $d\alpha = d\beta = 0$ on $\Lambda^2\mathcal{H}$. Frobenius' theorem shows that $\mathcal{H}$ is integrable.

We now express this in connection notation. Let $(X_1, X_2) = (X, Y)$ and $G_{ij} = g(X_i, X_j)$. In the original angular coordinates $\theta_0^i = \arg z_i$, every T^2-invariant metric on M° can be written as

$$\begin{aligned} g &= h + G_{ij}(d\theta_0^i + A^i)(d\theta_0^j + A^j), \\ A^i &= A^i_\mu \, dx^\mu, \end{aligned}$$

where x^μ are local coordinates on B° and the one-forms A^i on the base are pulled back to M°. The orthogonal torus connection has components

$$\eta^i := d\theta_0^i + A^i = G^{ij} g(X_j, \cdot), \qquad \eta^i(X_j) = \delta^i_j.$$

They form the dual coframe to (X_1, X_2) on the vertical distribution, extended to vanish on $\mathcal{H}$. The metric dual forms satisfy

$$\alpha = G_{11}\eta^1 + G_{12}\eta^2, \qquad \beta = G_{12}\eta^1 + G_{22}\eta^2.$$

Writing $A = A^i e_i$ in a basis of the Lie algebra, the general curvature formula is $F = dA + \frac{1}{2}[A \wedge A]$. The Lie algebra of T^2 has zero bracket, so $F^i = dA^i$. The horizontal lifts satisfy

$$U_\mu = \partial_{x^\mu} - A^i_\mu X_i, \qquad [U_\mu, U_\nu] = -F^i_{\mu\nu} X_i.$$

Thus the integrability proved above gives $F^i = 0$.

The original circle-valued angles trivialize the principal region, so the A^i are globally defined on B°. Since B° is simply connected, $dA^i = 0$ implies $A^i = df^i$ for smooth functions f^i on B°. Define adapted angles by $\theta^i = \theta_0^i + f^i$. Then $d\theta_0^i + A^i = d\theta^i$, and the new connection potentials are $A^i - df^i = 0$. Consequently,

$$g = h + G_{ij}\, d\theta^i d\theta^j, \qquad \partial_{\theta^i} = X_i.$$

Here the adapted angles θ^i take values in $\mathbb{R}/2\pi\mathbb{Z}$.

Note that G_{ij} is a globally defined matrix-valued function on B° : for any $q \in B^\circ$ and any $p \in \pi^{-1}(q)$, $G_{ij}(q) := g_p(X_i(p), X_j(p))$ is independent of the choice of p. Set

$$\rho := \sqrt{\det G}.$$

Let K be the Gaussian curvature of h. Let $\mathcal{V} = \mathcal{H}^\perp$ be the vertical distribution.

By the Koszul formula, for any basic horizontal fields U, V, the connection formulas are

$$\nabla^g_U V = \nabla^h_U V,$$
$$\nabla^g_U X_i = \nabla^g_{X_i} U = \frac{1}{2} G^{jk} U(G_{ki}) X_j,$$
$$\nabla^g_{X_i} X_j = -\frac{1}{2} \nabla^h G_{ij}.$$

Let μ, ν denote horizontal coordinate indices, write $G_\mu = \partial_\mu G$, and let $G_{ij;\mu\nu}$ denote the h-Hessian of the scalar component G_{ij}. The curvature tensor relation is

$$\begin{aligned}
R_{ijkl} &= \frac{1}{4} \left(\langle dG_{il}, dG_{jk} \rangle_h - \langle dG_{ik}, dG_{jl} \rangle_h \right), \\
R_{\mu i \nu j} &= -\frac{1}{2} G_{ij;\mu\nu} + \frac{1}{4} (G_\nu G^{-1} G_\mu)_{ij}, \\
R_{\mu\nu ij} &= \frac{1}{4} \left(G_\nu G^{-1} G_\mu - G_\mu G^{-1} G_\nu \right)_{ij}, \\
R_{\mu\nu\kappa\lambda} &= K(h_{\mu\kappa} h_{\nu\lambda} - h_{\mu\lambda} h_{\nu\kappa}), \\
R_{\mu\nu\kappa i} &= R_{\mu ijk} = 0,
\end{aligned}$$

the Ricci curvature relation is

$$\begin{aligned}
\mathrm{Ric}_{ij} &= -\frac{1}{2} \Delta_h G_{ij} + \frac{1}{2} h^{\mu\nu} \left(G_\mu G^{-1} G_\nu \right)_{ij} - \frac{1}{2} \langle d \log \rho, dG_{ij} \rangle_h, \\
\mathrm{Ric}_{\mu\nu} &= K h_{\mu\nu} - \nabla^2_{\mu\nu} (\log \rho) - \frac{1}{4} \mathrm{tr}(G^{-1} G_\mu G^{-1} G_\nu), \\
\mathrm{Ric}_{\mu i} &= 0,
\end{aligned}$$

and the scalar curvature relation is

$$R_g = 2K - 2\Delta_h (\log \rho) - |d \log \rho|^2_h - \frac{1}{4} h^{\mu\nu} \mathrm{tr}(G^{-1} G_\mu G^{-1} G_\nu).$$

The vertical part of Ricci curvature is equivalent to

$$\mathrm{Ric}^{\mathcal{V}} = -\frac{1}{2\rho} G \mathrm{div}_h \left(\rho G^{-1} dG \right),$$

whose trace gives

$$\mathrm{tr}(G^{-1} \mathrm{Ric}^{\mathcal{V}}) = -\frac{\Delta_h \rho}{\rho}.$$

Using the Einstein equation $\mathrm{Ric}^{\mathcal{V}} = 3G$, for $H := \rho^{-1} G$ with $\det H = 1$, we obtain

$$\Delta_h \rho = -6\rho, \quad \mathrm{div}_h (\rho H^{-1} dH) = 0.$$

Expanding this matrix equation, we obtain

$$\mathrm{div}_h (\rho dH_{12}) - \rho e H_{12} = 0,$$

where

$$e := |dH_{12}|^2_h - \langle dH_{11}, dH_{22} \rangle_h = |dH_{12} - H_{12} d \log H_{11}|^2_h + |d \log H_{11}|^2_h \geq 0.$$

We now use this equation of H_{12} together with its boundary values to conclude that $H_{12} \equiv 0$. Near the boundary arc corresponding to $z_i = 0$, consider small neighborhoods as the orbit space of $r_i = |z_i| \ll 1$. Then $X_i = r_i E_i$ with $0 < c \leq |E_i|_g \leq C$ for some positive constants $c, C > 0$, which can be shown by the fact that

$|E_i|_{g_0} = 1$ for the round metric and the equivalence between g and g_0. The fields E_1, E_2 are orthonormal for the round comparison metric, so their entire g-Gram matrix is bounded below by cI, and its determinant is at least c^2. At an ordinary axis the collapsing-circle half-turn reverses its normal plane and fixes the tangent plane, forcing the limiting cross term to vanish; at a corner the two distinct torus rotation planes are orthogonal. Smoothness then gives $g(E_1, E_2) \to 0$ along every approach to the boundary. Thus

$$H_{12} = \frac{g(E_1, E_2)}{\sqrt{g(E_1, E_1)g(E_2, E_2) - g(E_1, E_2)^2}} \to 0 \text{ as } r_i \to 0.$$

Thus, by defining H_{12} as zero on the boundary ∂B, we obtain

$$H_{12} \in C^\infty(B^\circ) \cap C^0(B), \quad H_{12}|_{\partial B} = 0.$$

By the strong maximum principle, we have $H_{12} \equiv 0$. Thus, $G_{12} \equiv 0$ in B°, and, writing $(\phi, \theta) = (\theta^1, \theta^2)$ for the adapted angles above, we can write

$$g = h + a^2(d\phi)^2 + b^2(d\theta)^2,$$

for some smooth functions $a, b > 0$ on B°.

2.2. **Curvature upper bound in the regular part.** For the diagonal metric obtained above, the curvature formulas simplify. On the principal region M°,

$$g = h + a^2(d\phi)^2 + b^2(d\theta)^2 \text{ on } M^\circ,$$

with $a, b > 0$ on B° and $b = 0$ on Γ_1 and $a = 0$ on Γ_2. Recall $\rho = ab$. Let E_ϕ, E_θ be vertical orthonormal fields, and U be any unit horizontal field. Then

$$\begin{aligned}
\sec_g|_{\mathcal{H}} &= K, \\
\sec_g(U, E_\phi) &= -\frac{(\nabla_h^2 a)(U, U)}{a}, \\
\sec_g(U, E_\theta) &= -\frac{(\nabla_h^2 b)(U, U)}{b}, \\
\sec_g(E_\phi, E_\theta) &= -\frac{\langle da, db\rangle_h}{ab},
\end{aligned}$$

and

$$\begin{aligned}
\mathrm{Ric}_g|_{\mathcal{H}} &= Kh - \frac{\nabla_h^2 a}{a} - \frac{\nabla_h^2 b}{b}, \\
\mathrm{Ric}_g(E_\phi, E_\phi) &= -\frac{\Delta_h a}{a} - \frac{\langle da, db\rangle_h}{ab}, \\
\mathrm{Ric}_g(E_\theta, E_\theta) &= -\frac{\Delta_h b}{b} - \frac{\langle da, db\rangle_h}{ab}, \\
\mathrm{Ric}_g(E_\phi, E_\theta) &= \mathrm{Ric}_g(E_\phi, U) = \mathrm{Ric}_g(E_\theta, U) = 0,
\end{aligned}$$

and

$$R_g = 2K - \frac{2\Delta_h a}{a} - \frac{2\Delta_h b}{b} - \frac{2\langle da, db\rangle_h}{ab}.$$

Within this subsection, unmarked derivatives and tensor norms are taken with respect to h.

Using $\mathrm{Ric}_g = 3g$, we have

$$\frac{\nabla^2 a}{a} + \frac{\nabla^2 b}{b} = (K-3)h,$$
$$\frac{\Delta a}{a} + \frac{\langle da, db\rangle_h}{ab} = \frac{\Delta b}{b} + \frac{\langle da, db\rangle_h}{ab} = -3.$$

So

$$K = -\frac{\langle da, db\rangle_h}{ab}, \quad \Delta\rho = -6\rho.$$

Applying the Bochner formula to (B°, h),

$$\Delta\langle da, db\rangle = \langle d\Delta a, db\rangle + \langle da, d\Delta b\rangle + 2\langle\nabla^2 a, \nabla^2 b\rangle + 2K\langle da, db\rangle.$$

Therefore,

$$\Delta K + 3\langle d\log\rho, dK\rangle = 3(K-1)(K+3) + \frac{1}{2}\left|a^{-1}\nabla^2 a - b^{-1}\nabla^2 b\right|^2.$$

We prove $K \le 1$ on B° by a cutoff argument.

First, by the curvature relation, we know that $|K| \le \max|\mathrm{sec}_g| \le C(g)$ in B° for some constant $C(g)$ depending on g. Second, $\rho = |X \wedge Y|_g > 0$ is well defined on M° and can be extended as a global Lipschitz function on M so that $|\nabla_g\rho| \le C(g)$ and $\rho = 0$ precisely on $M \setminus M^\circ$.

Note that for any T^2-invariant function f on M°,

$$\Delta_g f = \Delta_h f + \langle d\log\rho, df\rangle_h.$$

We can rewrite the equation of K as

$$\begin{aligned}\mathrm{div}_g(\rho^2\nabla_g K) &= \rho^2\left(3(K-1)(K+3) + \frac{1}{2}\left|a^{-1}\nabla_h^2 a - b^{-1}\nabla_h^2 b\right|_h^2\right)\\ &\ge 3\rho^2(K-1)(K+3).\end{aligned}$$

Choose a cutoff function η so that $\eta = 0$ on $[0,1]$, $\eta = 1$ on $[2,\infty)$ and $0 \le \eta' \le 2$. For $0 < \epsilon \ll 1$, define

$$\chi_\epsilon := \eta\left(\frac{\rho}{\epsilon}\right).$$

Then $\mathrm{supp}\chi_\epsilon \subset M^\circ$. Set $w = (K-1)_+$. Multiplying this inequality by $\chi_\epsilon^2 w$, and integrating by parts, we obtain

$$\begin{aligned}3\int_{M^\circ}\chi_\epsilon^2\rho^2 w^2(K+3) &\le -\int_{M^\circ}\langle\rho^2\nabla_g K, \nabla_g(\chi_\epsilon^2 w)\rangle_g\\ &= -\int_{M^\circ}\rho^2\chi_\epsilon^2|\nabla_g w|^2 - 2\int_{M^\circ}\rho^2\chi_\epsilon w\langle\nabla_g w, \nabla_g\chi_\epsilon\rangle,\end{aligned}$$

which gives

$$\begin{aligned}\frac{1}{2}\int_{M^\circ}\rho^2\chi_\epsilon^2|\nabla_g w|^2 + 3\int_{M^\circ}\chi_\epsilon^2\rho^2 w^2(K+3) &\le 2\int_{M^\circ}\rho^2 w^2|\nabla_g\chi_\epsilon|^2\\ &\le C(g)\int_{\{\epsilon\le\rho\le 2\epsilon\}} w^2\\ &\le C(g)\mathrm{Vol}(\{\epsilon\le\rho\le 2\epsilon\}) \to 0,\end{aligned}$$

as $\epsilon \to 0$. On the other hand, $\chi_\epsilon \to 1$ as $\epsilon \to 0$, so we obtain

$$\int_{M^\circ \cap \{K>1\}} \rho^2 w^2 (K+3) = 0,$$

which proves the claim that $K \le 1$ in B°.

2.3. **Convex exhaustion criterion.**

Proposition 2.1. *Let (Ω, h) be a connected smooth Riemannian manifold without boundary, not necessarily complete. Suppose $\sec_h \le 1$. Assume that there exists $F \in C^\infty(\Omega)$ such that:*

$$\{F \le r\} \text{ is compact for every } r \in \mathbb{R}, \quad \nabla_h^2 F > 0.$$

Let d_h denote the induced length distance on (Ω, h), that is, $d_h(x,y) = \inf\{L_h(\gamma) : \gamma \subset \Omega$ is a smooth curve between x and $y\}$. Then both (Ω, d_h) and its metric completion are CAT(1).

Proof. Fix $p_0 \in \Omega$. Choose regular values $r_j \to \infty$ so that $p_0 \in \{F < r_1\}$. Let C_j be the connected component of $\{F \le r_j\}$ containing p_0, and let d_j be the induced length metric on C_j. Then $C_j \subset C_{j+1}$ and we have an exhaustion $\Omega = \cup_j C_j$.

Each C_j is locally geodesically convex. Indeed, for any $p \in C_j$, we can find a sufficiently small strongly convex neighborhood $V \subset \Omega$. By the curvature assumption, we can assume (V, d_h) is CAT(1). For any $x, y \in V \cap C_j$, let γ be the constant-speed minimizing h-geodesic. Since $\nabla_h^2 F > 0$, we know $F(\gamma(t)) \le (1-t)F(x) + tF(y) \le r_j$. Therefore, $V \cap C_j$ is geodesically convex in V, and $d_j(x,y) = d_h(x,y)$ for all such $x, y \in V \cap C_j$. Thus (C_j, d_j) is a compact, locally CAT(1) length space. Each of its local geodesics is locally an ambient h-geodesic, by the local metric agreement. Along a nonconstant closed such geodesic γ, strict convexity would give $(F\circ\gamma)'' > 0$, contradicting periodicity. Hence there are no closed local geodesics. The compact globalization criterion makes (C_j, d_j) globally CAT(1) [1, Corollary 3.3.4].

For $j' \ge j$, a minimizing $d_{j'}$-geodesic between points of C_j is locally an ambient geodesic. Convexity of F keeps it in $\{F \le r_j\}$, hence in the same component C_j. Thus $d_{j'} = d_j$ on C_j. Every ambient comparison curve lies in some C_k, so $d_h = d_j$ on C_j as well. For any geodesic triangle in (Ω, d_h), one can choose large enough j so that it lies in C_j, and thus (Ω, d_h) is CAT(1). Finally, by Bridson–Haefliger [3, Corollary II.3.11], metric completion preserves CAT(1). □

In our case, $\{-\log\rho \le r\} \Subset B^\circ$ is compact and gives an exhaustion of B°. Moreover,

$$\begin{aligned}\nabla_h^2(-\log\rho) &= (3-K)h + d\log a \otimes d\log a + d\log b \otimes d\log b \\ &\ge 2h > 0.\end{aligned}$$

Consequently, both (B°, d_h) and its completion $(\widehat{B^\circ, d_h})$ are CAT(1).

Let (B, d_B) denote the orbit distance, that is,

$$d_B(\pi(p), \pi(q)) = \min_{\tau \in T^2} d_g(p, \tau(q)).$$

We claim that $(B, d_B) = (\widehat{B^\circ, d_h})$, so Proposition 2.1 makes (B, d_B) a CAT(1) space. It suffices to show that $(B^\circ, d_h) \subset (B, d_B)$ is an isometric inclusion.

Projection of curves gives $d_B \le d_h$. To prove the reverse inequality, take two points of B° and choose representatives $p, q \in M^\circ$ so that $d_B(\pi(p), \pi(q)) = d_g(p, q)$

and let $\gamma \subset (M, g)$ be a minimizing geodesic between p and q. Then $\gamma(s) = (z_1(s), z_2(s), t(s))$ may have intersection with singular orbits $\{(z_1, z_2, t) : z_1 z_2 = 0\}$. We can perform a slight perturbation so that each $z_i(s) \in \mathbb{C}$ avoids $0 \in \mathbb{C}$ and therefore obtain perturbations $\gamma_n \subset M^\circ$ with the same endpoints so that $\gamma_n \to \gamma$ in C^1. The conclusion then follows from $d_h(\pi(p), \pi(q)) \le L_h(\pi(\gamma_n)) \le L_g(\gamma_n) \to d_g(p, q)$.

2.4. **Maximal diameter and rigidity.**

Lemma 2.2. *Let a compact group G act isometrically on a closed Riemannian manifold (M, g). Let F be a connected component of $\mathrm{Fix}(H)$, for some subgroup $H \subset G$, and assume F is G-invariant. Give F its induced Riemannian metric and F/G its intrinsic quotient distance. Then $F/G \subset M/G$ locally preserves distances.*

Proof. Take two orbit points $x, y \in F/G$ whose distance in M/G is less than $\mathrm{inj}(M, g)$, and take $p, q \in F$ so that $d_{M/G}(x, y) = d_g(p, q)$. Let $\gamma \subset (M, g)$ be the unique minimizing geodesic between p and q. So for any $h \in H$, $h \circ \gamma = \gamma$, thus $\gamma \subset F$ and $d_F(p, q) = d_g(p, q)$. Therefore, $d_{F/G}(x, y) \le d_{M/G}(x, y)$. The opposite inequality always holds. □

In our case, let $F_1 = \{z_2 = 0\}$ and $F_2 = \{z_1 = 0\}$ so that $\pi(F_i) = \Gamma_i$, $H_1 = \{1\} \times S^1$ and $H_2 = S^1 \times \{1\}$. Then $F_i = \mathrm{Fix}(H_i)$, so Lemma 2.2 makes each $\Gamma_i = F_i/T^2 \simeq [0, \ell_i] \subset B$ a local geodesic.

By Bridson–Haefliger [3, Proposition II.1.4], inside a global CAT(1) space, every local geodesic of length less than π is minimizing, and two points at distance less than π have a unique minimizing geodesic. Therefore, $\max\{\ell_1, \ell_2\} \ge \pi$. An arc of length at least π has initial subarcs of every length $s < \pi$, each minimizing by the same CAT property. Letting $s \uparrow \pi$ gives $\mathrm{diam}(B) \ge \pi$.

Since π is 1-Lipschitz, together with Bonnet-Myers, we have

$$\pi \le \mathrm{diam}(B) \le \mathrm{diam}(M, g) \le \pi.$$

Thus, $\mathrm{diam}(M, g) = \pi$. Cheng's maximal-diameter rigidity [6] proves Theorem 1.1.

3. Complete Ricci-flat metrics on $\mathbb{R}^4$

This section proves Theorem 1.2.

3.1. **Model spaces.**

The two models in Theorem 1.2 admit the following explicit expressions. On $\mathbb{C}^2$, write

$$\zeta_1 = x e^{i\theta^1}, \quad \zeta_2 = y e^{i\theta^2}, \quad x, y \ge 0, \quad \theta^j \in \mathbb{R}/(2\pi\mathbb{Z}).$$

Green functions in these models are functions on the four-manifold, normalized by $-\Delta_g G_p = 4\pi^2 \delta_p$. The torus translates the two angles, with generators $X_j = \partial_{\theta^j}$. Its quotient is the quadrant with coordinates $\{(x, y) : x, y \ge 0\}$. Set $r^2 = x^2 + y^2$. The Hodge-star computations below are on the smooth quotient interior $x, y > 0$; the displayed scalar formulas extend continuously to the axes.

3.1.1. *Euclidean space.* The metric, quotient metric and orbit Gram matrix are

$$g_0 = dx^2 + dy^2 + x^2(d\theta^1)^2 + y^2(d\theta^2)^2,$$
$$h_0 = dx^2 + dy^2, \quad G_0 = \begin{pmatrix} x^2 & 0 \\ 0 & y^2 \end{pmatrix}.$$

Thus

$$\rho = \sqrt{\det G_0} = xy.$$

Since $*_{h_0} dx = dy, *_{h_0} dy = -dx$,

$$d\tau = - *_{h_0} d(xy) = xdx - ydy, \quad \tau = \frac{x^2 - y^2}{2}.$$

For

$$\xi := \sqrt{\rho^2 + \tau^2} = \frac{x^2 + y^2}{2},$$

we have

$$x = \sqrt{\xi + \tau}, \ y = \sqrt{\xi - \tau}.$$

Thus (ρ, τ) identifies the closed quotient with the half-plane $\rho \geq 0$, conformally on its interior; equivalently $\tau + i\rho = \frac{1}{2}(x + iy)^2$, and

$$d\rho^2 + d\tau^2 = 2\xi h_0.$$

The normalized Green function of g_0 at the origin is

$$G_p = \frac{1}{2\xi} = \frac{1}{r^2},$$

and the full Weyl expression is

$$g_0 = \frac{d\rho^2 + d\tau^2}{2\xi} + (\xi + \tau)(d\theta^1)^2 + (\xi - \tau)(d\theta^2)^2.$$

3.1.2. *Taub–NUT.* Set

$$\eta = x^2 d\theta^1 + y^2 d\theta^2, \quad f(r) = 1 + \frac{1}{4} r^2, \quad b(r) = \frac{1}{4} + \frac{1}{4f(r)}.$$

Then the Taub–NUT metric can be written as

$$g_{\mathrm{TN}} = f g_0 - b\eta^2.$$

The metric has eigenvalues f, f, f, f^{-1} relative to g_0. The quotient metric and orbit Gram matrix are

$$h_{\mathrm{TN}} = f(dx^2 + dy^2), \quad G_{\mathrm{TN}} = \begin{pmatrix} fx^2 - bx^4 & -bx^2y^2 \\ -bx^2y^2 & fy^2 - by^4 \end{pmatrix}.$$

Since $f - br^2 = f^{-1}$,

$$\rho_{\mathrm{TN}} = \sqrt{\det G_{\mathrm{TN}}} = xy.$$

Since $h_{\mathrm{TN}} = f h_0$ and the Hodge star on one-forms is conformally invariant in dimension two,

$$d\tau = - *_{h_{\mathrm{TN}}} d(xy) = xdx - ydy, \quad \tau = \frac{x^2 - y^2}{2}, \quad \xi = \frac{x^2 + y^2}{2}.$$

The same half-plane coordinates therefore give

$$h_{\mathrm{TN}} = \left(\frac{1}{2\xi} + \frac{1}{4}\right)(d\rho^2 + d\tau^2).$$

Here r is the coordinate radius on $\mathbb{C}^2$. The normalized Green function on $(\mathbb{R}^4, g_{\mathrm{TN}})$ is again r^{-2}.

3.2. **Action reduction.** Let M^4 be a smooth 4-manifold homeomorphic to $\mathbb{R}^4$, which admits an effective smooth T^2-action. By Orlik–Raymond, the T^2-symmetry forces its smooth structure to be standard, and up to a diffeomorphism of $\mathbb{R}^4$ and an automorphism of T^2-action, we can assume that the effective T^2-action is given by the standard action:

$$\left(e^{is_1}, e^{is_2}\right) \cdot (z_1, z_2) = (e^{is_1} z_1, e^{is_2} z_2).$$

To see this, extend the action continuously to $\hat{M} = M \cup \{\infty\} \simeq S^4$ by fixing ∞. Apply the continuous-action appendix of Orlik–Raymond with $A = \varnothing$, since $\hat{M} \simeq S^4$ is an orientable integral cohomology four-manifold. Since $H_1(\hat{M}; \mathbb{Z}) = 0$, their classification gives a disk quotient with no exceptional finite-isotropy orbits. The fixed-point count is $\chi(\hat{M}) = 2$. Its two boundary arcs have primitive isotropy generators forming an integral basis. Deleting ∞ leaves a quadrant with one corner and two rays.

This compactification is used only for the topological orbit data. On the original smooth M, the smooth slice models make the quotient a smooth surface with corners. Relative uniqueness of smooth structures on surfaces identifies it with the standard quadrant, preserving the labels and corner chart [12, Theorem B and relative-boundary version]; smooth action reconstruction then gives the asserted equivariant diffeomorphism. This reconstruction applies also to noncompact manifolds [18, §§1.11–1.12, p. 537, and Appendix].

Now let g be a smooth complete Ricci-flat metric preserved by the action. Let $\pi : M \to X$ be the quotient map and $p \in M$ the unique torus fixed point, projecting to the corner. Write $X^\circ \simeq (0, \infty)^2$ and $M^\circ = \pi^{-1}(X^\circ)$. Let X_1, X_2 be the primitive 2π-periodic Killing generators and α_i their metric dual forms.

Since X° is contractible, the free orbit bundle is trivial, and in standard angular coordinates θ_0^i, a general T^2-invariant metric has the form

$$g = h + G_{ij}(d\theta_0^i + A^i)(d\theta_0^j + A^j).$$

The twist argument in Section 2.1 applies because both generators vanish at p. Thus $\mathcal{H} = \ker \alpha_1 \cap \ker \alpha_2$ is integrable. So $dA^i = 0$, and contractibility gives $A^i = df^i$ globally, and the change $\theta^i = \theta_0^i + f^i$ modulo 2π gives

$$g = h + G_{ij} d\theta^i d\theta^j \text{ on } M^\circ.$$

Let $\rho = \sqrt{\det G}$ be the fiber density function. Then $\rho > 0$ in X° and $\rho = 0$ on ∂X.

3.3. **Global Weyl coordinates.** Using the Ricci-flat equation $\mathrm{Ric}_g = 0$, we obtain

$$\Delta_h \rho = 0.$$

We can construct the harmonic conjugate as follows. Fix an orientation of the smooth surface (X°, h). The one-form $- *_h d\rho$ is closed, since $d(- *_h d\rho) =$

$-(\Delta_h\rho)dV_h = 0$. As X° is simply connected, there is a smooth function $\tau : X^\circ \to \mathbb{R}$, unique up to an additive constant, with $d\tau = -*_h d\rho$. Thus,

$$\Delta_h\tau = 0, \quad \langle d\rho, d\tau\rangle_h = 0, \quad |d\tau|_h = |d\rho|_h.$$

The goal is to show that a continuous extension of (ρ, τ) gives a global coordinate of X.

Lemma 3.1. *The lift $\tilde\tau = \tau \circ \pi$ extends smoothly and harmonically to M, and can be normalized by $\tilde\tau(p) = 0$. Also, $\tilde\tau$ descends to a continuous function on X, which is still denoted by τ. Define $\xi := \sqrt{\rho^2 + \tau^2}$. Near p, we have the expansion*

$$\xi = \frac{1}{2}d^2 + O(d^4),$$

where $d = d_g(p, \cdot)$.

Proof. On M°, we have

$$\Delta_g\tilde\tau = (\Delta_h\tau + \langle d\log\rho, d\tau\rangle_h) \circ \pi = 0, \quad |\nabla\tilde\tau|_g = |\nabla\tilde\rho|_g,$$

where $\tilde\rho = \rho \circ \pi = |X_1 \wedge X_2|_g$. So $|\nabla\tilde\tau|_g$ is locally bounded.

The singular set is the union of two smooth codimension-two fixed surfaces meeting at p. In a sufficiently small geodesically convex ball U, a geodesic between principal points can be approximated, with fixed endpoints and converging lengths, by smooth curves avoiding both surfaces. Indeed, a transverse curve misses each codimension-two stratum and the isolated point. Integrating the locally bounded gradient along these curves gives

$$|\tilde\tau(x) - \tilde\tau(y)| \le L_U d_g(x, y), \qquad x, y \in U \cap M^\circ.$$

Thus $\tilde\tau$ extends uniquely as a locally Lipschitz function on M, and belongs to $W^{1,2}_{\mathrm{loc}}(M)$. Continuity preserves torus invariance, so the extension descends continuously to X.

As in the S^4-case, let $\chi_\epsilon = \eta(\epsilon^{-1}\rho)$ be a cutoff function for some η so that $\eta = 0$ on $[0, 1]$, $\eta = 1$ on $[2, \infty)$ and $0 \le \eta' \le 2$. For any test function $\varphi \in C^\infty_c(M)$, we have

$$\int_M \chi_\epsilon\langle\nabla^g\tilde\tau, \nabla^g\varphi\rangle_g\, dV_g = -\int_M \varphi\langle\nabla^g\tilde\tau, \nabla^g\chi_\epsilon\rangle_g\, dV_g = 0.$$

Here the cutoff uses the lift of ρ. Orthogonality of $d\tau$ and $d\rho$ makes the last integral zero. The singular set has zero Riemannian volume, so dominated convergence proves weak harmonicity on M. Elliptic regularity makes $\tilde\tau$ smooth.

Normalize $\tilde\tau(p) = 0$. The action reduction has fixed a primitive torus basis whose tangent representation at p consists of two standard rotation planes. These planes are orthogonal for the invariant inner product g_p. Choose orthonormal coordinates on them and use the exponential map, which intertwines this linear action with the given action. The resulting local geodesic normal coordinates (u_1, u_2, u_3, u_4) satisfy $g_{\alpha\beta} = \delta_{\alpha\beta} + O(|u|^2)$ and the exact formulas

$$X_1 = -u_2\partial_{u_1} + u_1\partial_{u_2}, \quad X_2 = -u_4\partial_{u_3} + u_3\partial_{u_4}.$$

Set

$$x_p^2 = u_1^2 + u_2^2, \qquad y_p^2 = u_3^2 + u_4^2, \qquad d^2 = d_g(p, \cdot)^2 = |u|^2 = x_p^2 + y_p^2.$$

In this local computation, abbreviate $x = x_p$ and $y = y_p$; global action radii will be specified separately.

Since G_{ij} is T^2-invariant, we can focus on one slice $(u_1, u_2, u_3, u_4) = (x, 0, y, 0)$. Then

$$\begin{aligned} G_{11} &= \langle X_1, X_1\rangle_g = x^2 g_{22} = x^2(1 + O(d^2)), \\ G_{22} &= \langle X_2, X_2\rangle_g = y^2 g_{44} = y^2(1 + O(d^2)), \\ G_{12} &= xy g_{24} = xy \cdot O(d^2). \end{aligned}$$

In fact, for later use, we have better asymptotics of G_{12}. Note that $g_{24}(0, 0, y, 0) = g_{24}(x, 0, 0, 0) = 0$. To see this, one can consider the isometric rotation by angle π in the plane (u_3, u_4), which fixes the point $m = (x, 0, 0, 0)$, so that $g(\partial_{u_2}, \partial_{u_4})(m) = g(\partial_{u_2}, -\partial_{u_4})(m) = 0$. So $g_{24}(x, 0, y, 0) = xy\hat{g}_{24}(x, 0, y, 0)$ for some smooth function $\hat{g}_{24}$, and

$$G_{12} = x^2 y^2 \hat{g}_{24} = x^2 y^2 \cdot O(1).$$

Around p, we have the expansion

$$\rho = xy(1 + O(d^2)).$$

Torus invariance, harmonicity, and $\tilde{\tau}(p) = 0$ give the Taylor expansion

$$\tilde{\tau}(u) = Bx^2 - By^2 + O(d^4).$$

The gradient identity gives $4B^2 = 1$. Choose the orientation of X° so that $B = 1/2$; reversing the orientation reverses τ. By the definition, we have

$$\xi = \frac{1}{2}d^2 + O(d^4).$$

□

Lemma 3.2. *Let E_+ and E_- be the boundary rays of X, characterized near the corner by $y_p = 0$ and $x_p = 0$ respectively as above. Orient both rays away from the corner. Then τ is strictly increasing on E_+ and strictly decreasing on E_-. In particular, ξ vanishes only at the corner. At every ordinary axis point, (ρ, τ) is a local diffeomorphism of closed half-plane neighborhoods.*

Proof. Set $W = X_1 \wedge X_2$. Then $|W|_g^2 = \rho^2 \circ \pi$. On M°, we have

$$\Delta_g |W|_g^2 = 4|\nabla\rho|_h^2 \circ \pi = 4|\nabla\tilde{\rho}|_g^2 = 4|\nabla\tilde{\tau}|_g^2.$$

The first and last expressions are smooth on M, so their equality holds everywhere.

Let F be the fixed surface of X_1-generated action and $q \in F$ be an ordinary axis point. Then $X_1(q) = 0$ and $X_2(q) \neq 0$ is tangent to F. Since $W(q) = 0$,

$$(\Delta_g |W|_g^2)(q) = 2|\nabla W|_g^2(q) = 4|X_2|_g^2(q).$$

Indeed, $A := (\nabla X_1)(q)$ vanishes on T_qF and generates a rotation on N_qF by the Killing equation. Its weight is one in absolute value: a higher weight would give nontrivial finite isotropy at nearby principal points, whereas X_1 is primitive of period 2π and the principal action is free. Choose an orthonormal frame so that $e_1, e_2 \in N_qF$ with $Ae_1 = e_2, Ae_2 = -e_1$, and $e_3, e_4 \in T_qF$, then $(\nabla_v W)(q) = Av \wedge X_2(q)$. Consequently, $|\nabla W|_g^2(q) = |e_2 \wedge X_2(q)|^2 + |e_1 \wedge X_2(q)|^2 = 2|X_2|_g^2(q)$.

In particular, $|\nabla\tilde{\tau}|_g(q) > 0$ at any ordinary axis point q in F. Note that $\nabla\tilde{\tau}(q) \in T_qF$. To see this, let σ be the angle π rotation generated by X_1, then $\sigma(q) = q$ and $d_q\sigma = -\mathrm{Id}$ on N_qF. For any $v \in N_qF$, we have $d_q\tilde{\tau}(v) = d_q(\tilde{\tau} \circ \sigma)(v) = d_q\tilde{\tau}(-v)$, which implies that $d_q\tilde{\tau}(v) = 0$. Similarly, $\nabla\tilde{\tau}(q) \perp X_2(q)$. This shows that $\tilde{\tau}|_F$ descends to $\tau|_{E_\pm}$ with $d(\tau|_{E_\pm}) \neq 0$ at every ordinary boundary point. So $\tau|_{E_\pm}$ is

either strictly monotone increasing or decreasing, with the sign determined by the corner expansion. The same calculation with the indices exchanged applies to the other ray.

To show the local diffeomorphism at any ordinary axis point, we use the global equivariant identification $M \simeq \mathbb{C}^2$. The Cartesian coordinates are $z_j = a_j + ib_j, x = |z_1|, y = |z_2|$ and $(x, y) \in [0, \infty)^2$ are global coordinates on X. Note the difference from the normal coordinates (x_p, y_p) we used before.

Fix $(0, y_0) \in E_-$ with $y_0 > 0$. Choose small enough $\epsilon > 0$ and consider the tubular neighborhood $0 \le x < \epsilon, |y - y_0| < \epsilon$, away from the corner. We can evaluate the metric on the slice $(a_1, b_1, a_2, b_2) = (a_1, 0, y, 0)$ with $|a_1| < \epsilon, |y - y_0| < \epsilon$, where $X_1 = a_1\partial_{b_1}, X_2 = y\partial_{b_2}$. So

$$G_{11} = a_1^2 g_{b_1 b_1}, \quad G_{12} = a_1 y g_{b_1 b_2}, \quad G_{22} = y^2 g_{b_2 b_2}.$$

Set $\tilde{B} = (g_{b_1 b_1} g_{b_2 b_2} - g_{b_1 b_2}^2)^{1/2} > 0$. On the signed slice the invariant function is $\tilde{\rho} = |a_1| y \tilde{B}$. The first-circle half-turn makes $g_{b_1 b_1}$ and $g_{b_2 b_2}$ even in a_1, and $g_{b_1 b_2}$ odd in a_1. Thus $\tilde{B}$ is smooth and even, and on the quotient we obtain

$$\rho = xyB(x^2, y) \text{ with } B(0, y) > 0, \quad \tau = \tau(x^2, y),$$

The previous computation shows that $\tau_y(0, y_0) \neq 0$. Extend the formulas $xyB(x^2, y)$ and $\tau(x^2, y)$ smoothly to signed x; the signed area formula agrees with ρ for $x \ge 0$. Their Jacobian at $(0, y_0)$ is $y_0 B(0, y_0)\tau_y(0, y_0) \neq 0$. The inverse function theorem, restricted to $x \ge 0$, proves the asserted local diffeomorphism.

□

Lemma 3.3. *The minimal positive Green function of (M^4, g) at p, normalized by $-\Delta_g G_p = 4\pi^2 \delta_p$, satisfies*

$$G_p = \frac{1}{2\tilde{\xi}}, \quad \tilde{\xi} := \xi \circ \pi.$$

It is smooth on $M \setminus \{p\}$ and has the local expansion $G_p \cdot d_g(p, \cdot)^2 \to 1$ around p.

Proof. Set $U = (2\tilde{\xi})^{-1}$. For a smooth function $v(\rho, \tau)$, the conjugacy identities give on M°

$$\Delta_g(v \circ \pi) = \left[|d\rho|_h^2 (v_{\rho\rho} + \rho^{-1} v_\rho + v_{\tau\tau})\right] \circ \pi.$$

The expression in parentheses vanishes for $v = (2\sqrt{\rho^2 + \tau^2})^{-1}$. Moreover, $\tilde{\xi}^2 = |X_1 \wedge X_2|_g^2 + \tilde{\tau}^2$ is smooth and positive off p, so U extends smoothly and harmonically across every ordinary axis. At p, the normal-coordinate expansion gives $U = d^{-2} + O(1)$ with the corresponding differentiated expansion. Its inward flux is therefore $4\pi^2$, proving $-\Delta_g U = 4\pi^2 \delta_p$.

Dirichlet Green functions on an exhaustion increase to a minimal positive Green function $G_p \le U$. Their difference extends harmonically across p (the pole cancels distributionally) and is nonnegative. Yau's harmonic Liouville theorem [21] makes $U - G_p = c$ for some $c \ge 0$. If $c > 0$, then $U \ge c$, hence $|\tilde{\tau}| \le \tilde{\xi} \le (2c)^{-1}$. The same Liouville theorem would make the globally harmonic $\tilde{\tau}$ constant, contradicting its nonzero quadratic term at p. Thus $c = 0$ and $G_p = U$. □

Proposition 3.4. *The map*

$$(\rho, \tau) : X \to \{(s, t) \in \mathbb{R}^2 : s \ge 0\}$$

is a proper homeomorphism and a conformal diffeomorphism on the interiors.

Proof. We first show this map is proper. By Li–Yau's Green-function estimate [16, Theorem 5.2], we know as $d_g(p,\cdot) \to \infty$, $G_p(\cdot) \to 0$, so that $\xi \circ \pi(\cdot) \to \infty$. Completeness makes the sublevel sets of $\xi \circ \pi$ compact on M. So (ρ, τ) is proper on the quotient.

Properness and the strict monotonicity of τ make τ map the two boundary rays bijectively onto the positive and negative real rays. Thus the boundary map is a homeomorphism. The map $w = \tau + i\rho$ is holomorphic on X°. Its nonempty image is open and closed, thus is the whole upper half-plane.

Fix an ordinary axis point q and a relatively open neighborhood U on which w is a diffeomorphism of half-plane neighborhoods. There is a neighborhood V of $w(q)$ such that $w^{-1}(V) \subset U$. Otherwise, choose $q_j \notin U$ with $w(q_j) \to w(q)$. Properness gives a convergent subsequence with limit $q_\infty \notin U$; since $\rho(q_\infty) = 0$, this contradicts the uniqueness of the boundary preimage of $w(q)$.

Choose an interior value in $V \cap w(U)$. Its only preimage lies in U, where the differential is invertible, so its local degree is one. A proper nonconstant holomorphic map has constant total degree, with each local multiplicity positive. Hence its degree is one everywhere, it has no branch points, and it is a conformal diffeomorphism of the interiors. Properness and the already bijective boundary map give the asserted homeomorphism of the closed quotients. □

3.4. **The sharp quotient bound.** By Proposition 3.4, (ρ, τ) are coordinates on X°. Put $\delta_2 = d\rho^2 + d\tau^2$ and $\Delta_2 = \Delta_{\delta_2}$. From this point onward, quotient scalar functions are identified with their invariant pullbacks when used on M. We retain tildes when explicitly distinguishing a function from its lift; metric subscripts specify all norms and differential operators. Applying Colding's sharp estimate [9, Theorem 3.1] to the function $b = G_p^{-\frac{1}{2}} = \sqrt{2\tilde{\xi}}$ on (M, g), we obtain

$$|\nabla b|_g \le 1.$$

On M°, we have

$$|\nabla b|_g^2 = \frac{|d\xi|_h^2}{2\xi} \circ \pi = \frac{|d\rho|_h^2}{2\xi} \circ \pi.$$

Note that

$$d\rho^2 + d\tau^2 = |d\rho|_h^2 h.$$

By Proposition 3.4, $|d\rho|_h \neq 0$, so we can write

$$h = e^{2\nu}(d\rho^2 + d\tau^2) \text{ on } X^\circ,$$

where $\nu = -\log |d\rho|_h$. Colding's estimate gives

$$2\xi \ge |d\rho|_h^2 = e^{-2\nu}.$$

Integrating the gradient estimate from p, where $\tilde{\xi} = 0$, gives $\tilde{\xi} \le \frac{1}{2} d_g(p,\cdot)^2$. Set $F := 2\xi e^{2\nu}$. Then $F \ge 1$ and

$$h = \frac{F}{2\xi}(d\rho^2 + d\tau^2).$$

3.5. **The harmonic-map equations.** The normalized orbit matrix $H = \rho^{-1}G$ takes values in the hyperbolic plane. Its weighted harmonic-map equation becomes the ordinary harmonic-map equation after introducing an auxiliary Euclidean three-space.

We first identify the space $\mathcal{P}$ of positive symmetric determinant-one (2×2)-matrices with the hyperbolic plane $\mathbb{H}^2$, equipped with the curvature -1 metric. For $H \in \mathcal{P}$, we have

$$T_H\mathcal{P} = \{V = V^T : \mathrm{tr}(H^{-1}V) = 0\},$$

and $\forall B, C \in T_H\mathcal{P}$,

$$\langle B, C\rangle_H := \frac{1}{2}\mathrm{tr}\left(H^{-1}BH^{-1}C\right).$$

Let D denote the Levi-Civita connection of the target metric and also its pullback along H. If i denotes a coordinate direction on the domain, $H_i = \partial_i H$, and V is a tangent matrix field along H, then

$$D_iV = \partial_iV - \frac{1}{2}(H_iH^{-1}V + VH^{-1}H_i).$$

To see the identification, one can take the coordinates

$$H = \frac{1}{u}\begin{pmatrix} u^2+v^2 & v \\ v & 1\end{pmatrix}, \quad u = H_{22}^{-1} > 0, \quad v = H_{22}^{-1}H_{12},$$

so that

$$g_{\mathcal{P}} = \frac{du^2 + dv^2}{u^2} = g_{\mathbb{H}^2}.$$

In the Weyl coordinates, recall the vertical Ricci equation

$$\begin{aligned} 0 &= \mathrm{div}_h\left(\rho H^{-1}dH\right) \\ &= e^{-2\nu}\left(\partial_\rho\left(\rho H^{-1}H_\rho\right) + \partial_\tau\left(\rho H^{-1}H_\tau\right)\right). \end{aligned}$$

That is, when $\rho > 0$,

$$H_{\rho\rho} + \rho^{-1}H_\rho - H_\rho H^{-1}H_\rho + H_{\tau\tau} - H_\tau H^{-1}H_\tau = 0.$$

In terms of the pullback connection, this is

$$D_\rho H_\rho + \rho^{-1}H_\rho + D_\tau H_\tau = 0.$$

We introduce an auxiliary $\mathbb{R}^3$ to simplify the arguments. Rotate the half plane $\{(\rho, \tau) : \rho \geq 0\}$ around $\rho = 0$, and let $\zeta = (\zeta_1, \zeta_2, \tau)$ denote the Cartesian coordinates. Define $\rho = \sqrt{\zeta_1^2 + \zeta_2^2}$, $\mathcal{A} = \{(0, 0, \tau) : \tau \in \mathbb{R}\}$ and $\Omega = \mathbb{R}^3 \setminus \mathcal{A}$. On Ω, write $(\zeta_1, \zeta_2) = (\rho\cos\varphi_{\mathrm{aux}}, \rho\sin\varphi_{\mathrm{aux}})$, and lift the map by

$$H(\zeta) = H(\rho, \tau) : \Omega \to \mathbb{H}^2.$$

The auxiliary Euclidean metric and volume are

$$\delta_3 = d\rho^2 + \rho^2 d\varphi_{\mathrm{aux}}^2 + d\tau^2, \quad dV_{\delta_3} = \rho d\rho d\varphi_{\mathrm{aux}} d\tau,$$

and for a function $f = f(\rho, \tau)$,

$$\Delta_{\delta_3} f = f_{\rho\rho} + \rho^{-1}f_\rho + f_{\tau\tau} \text{ on } \Omega.$$

Write $\Delta_3 = \Delta_{\delta_3}$. With the orthonormal frame $e_1 = \partial_\rho$, $e_2 = \rho^{-1}\partial_{\varphi_{\mathrm{aux}}}$, $e_3 = \partial_\tau$ on Ω, the tension field of $H : (\Omega, \delta_3) \to (\mathbb{H}^2, g_{\mathbb{H}^2})$ is

$$\begin{aligned}\mathcal{T}_{\delta_3}(H) &= \sum_\alpha \left(D_{e_\alpha}(dH(e_\alpha)) - dH(\nabla^{\delta_3}_{e_\alpha} e_\alpha) \right) \\ &= D_\rho H_\rho + \rho^{-1} H_\rho + D_\tau H_\tau = 0.\end{aligned}$$

For base indices $a, b \in \{\rho, \tau\}$, the identities $G = \rho H$ and $\mathrm{tr}(H^{-1}H_a) = 0$ give

$$\mathrm{tr}\left(G^{-1}G_a G^{-1}G_b\right) = 2(\partial_a \log\rho)(\partial_b \log\rho) + 2\langle H_a, H_b\rangle_{\mathbb{H}^2}.$$

Thus the horizontal Ricci-flat equation is

$$\mathrm{Ric}_h = \rho^{-1}\nabla_h^2\rho - \frac{d\rho \otimes d\rho}{2\rho^2} + \frac{1}{2}H^* g_{\mathbb{H}^2}.$$

On the other hand, for $h = e^{2\nu}\delta_2$, its Christoffel symbols are $\Gamma^c_{ab} = \delta^c_a\nu_b + \delta^c_b\nu_a - \delta_{ab}\delta^{cd}\nu_d$, so we have

$$\mathrm{Ric}_h = -(\Delta_2\nu)\delta_2, \quad (\nabla_h^2\rho)_{ab} = \begin{pmatrix} -\nu_\rho & -\nu_\tau \\ -\nu_\tau & \nu_\rho \end{pmatrix}_{ab}.$$

Thus we obtain

$$\nu_\rho = -\frac{1}{4\rho} + \frac{\rho}{4}\left(|H_\rho|^2_{\mathbb{H}^2} - |H_\tau|^2_{\mathbb{H}^2}\right), \quad \nu_\tau = \frac{\rho}{2}\langle H_\rho, H_\tau\rangle_{\mathbb{H}^2},$$

and

$$-\Delta_2\nu = \rho^{-1}\nu_\rho + \frac{1}{2}|H_\tau|^2_{\mathbb{H}^2}.$$

On the auxiliary space, $\xi = |\zeta|$ and $F = 2\xi e^{2\nu} \geq 1$. The scalar equations are

$$\Delta_3\nu = -\frac{1}{2}|H_\tau|^2_{\mathbb{H}^2}, \quad \Delta_3 \log F = \xi^{-2} - |H_\tau|^2_{\mathbb{H}^2}.$$

3.6. **The sharp Jacobi estimate.**

Proposition 3.5. *The norm of $J = H_\tau$, initially defined on Ω, extends to a locally Lipschitz, weakly subharmonic function on $\mathbb{R}^3 \setminus \{0\}$. This extension satisfies*

$$|J|_{\mathbb{H}^2} \leq \xi^{-1} \quad \text{on } \mathbb{R}^3 \setminus \{0\}.$$

Proof. Translations in τ preserve (Ω, δ_3), so $H(\zeta_1, \zeta_2, \tau + s)$ is harmonic for every s. Its variation field is $J = H_\tau$, a section of $H^*T\mathbb{H}^2$. Let $E_1 = \partial_{\zeta_1}$, $E_2 = \partial_{\zeta_2}$, $E_3 = \partial_\tau$, and $Y_i = dH(E_i)$. Thus $J = Y_3 = dH(E_3)$. Write $D_i = D_{E_i}$. In particular, $D_\tau = D_3$ differentiates a vector field along the target curve $t \mapsto H(\zeta_1, \zeta_2, t)$.

Since $[\partial_\tau, E_i] = 0$, we have

$$D_i J = D_\tau Y_i, \quad (D_\tau D_i - D_i D_\tau)V = \mathrm{R}_{\mathbb{H}^2}(J, Y_i)V.$$

Since the Cartesian frame is parallel, the harmonic map equation is $\sum_i D_i Y_i = 0$. Applying D_τ and using the preceding commutator yields the harmonic-map Jacobi equation

$$\begin{aligned}0 &= \sum_{i=1}^3 (D_i D_i J + R_{\mathbb{H}^2}(J, Y_i)Y_i) \\ &= \Delta^D J - \sum_{i=1}^3 \left(|Y_i|^2_{\mathbb{H}^2} J - \langle J, Y_i\rangle_{\mathbb{H}^2} Y_i\right).\end{aligned}$$

Thus

$$\begin{aligned}\frac{1}{2}\Delta_3|J|^2_{\mathbb{H}^2} &= |DJ|^2_{\delta_3,\mathbb{H}^2} + \langle \Delta^D J, J\rangle_{\mathbb{H}^2} \\ &= |DJ|^2_{\delta_3,\mathbb{H}^2} + \sum_{i=1}^{3}\left(|J|^2_{\mathbb{H}^2}|Y_i|^2_{\mathbb{H}^2} - \langle J, Y_i\rangle^2_{\mathbb{H}^2}\right) \\ &\geq |DJ|^2_{\delta_3,\mathbb{H}^2} \\ &\geq |\nabla^{\delta_3}|J|_{\mathbb{H}^2}|^2_{\delta_3}.\end{aligned}$$

Regularization gives

$$\Delta_3\sqrt{|J|^2_{\mathbb{H}^2}+\epsilon^2} \geq \frac{\epsilon^2}{(|J|^2_{\mathbb{H}^2}+\epsilon^2)^{3/2}}|DJ|^2_{\delta_3,\mathbb{H}^2} \geq 0,$$

taking $\epsilon \to 0$, we have

$$\Delta_3|J|_{\mathbb{H}^2} \geq 0 \text{ weakly on } \Omega.$$

We extend this weak subharmonic inequality through the ordinary axis. Fix an axis point with $\tau_0 \neq 0$, and exchange generators locally if necessary so that X_1 collapses. On the signed action slice from Lemma 3.2, write

$$g_{b_2b_2} = \ell(a_1^2, y), \qquad g_{b_1b_2} = a_1 k(a_1^2, y).$$

For $P = x^2$, this gives $G_{22} = y^2\ell(P,y)$ and $G_{12} = Pyk(P,y)$. The map

$$(P,y) \longmapsto (\rho^2,\tau) = (Py^2B(P,y)^2, \tau(P,y))$$

has Jacobian $y_0^2B(0,y_0)^2\tau_y(0,y_0) \neq 0$ at $(0,y_0)$. Its smooth inverse shows that

$$G_{22} = A(\rho^2,\tau) > 0, \qquad G_{12} = \rho^2 C(\rho^2,\tau).$$

Indeed $G_{12}/\rho^2 = k(P,y)/(yB(P,y)^2)$ is smooth there. Thus the target coordinates satisfy

$$u = \rho/A, \qquad v = \rho^2 C/A, \qquad |J|^2_{\mathbb{H}^2} = (A_\tau/A)^2 + \rho^2A^2\big((C/A)_\tau\big)^2.$$

Consequently $|J|$ is the norm of the smooth three-vector

$$\big(A_\tau/A,\ \zeta_1 A(C/A)_\tau,\ \zeta_2 A(C/A)_\tau\big),$$

and is locally Lipschitz through the axis. Test its weak inequality against $\varphi\eta(\rho/\epsilon)$ on a compact set avoiding zero. The extra gradient term is $O(\epsilon)$: $|\nabla^{\delta_3}\eta(\rho/\epsilon)| \leq C/\epsilon$, the tube has volume $O(\epsilon^2)$, and $|\nabla^{\delta_3}|J||$ is locally bounded. Passing to the limit proves

$$\Delta_3|J|_{\mathbb{H}^2} \geq 0 \quad \text{weakly on } \mathbb{R}^3\setminus\{0\}.$$

The square $|J|^2$ is smooth there; its subharmonicity extends by continuity.

Next, we show that $|J|_{\mathbb{H}^2} \to 0$ uniformly at infinity. Recall that $\Delta_3\nu = -\frac{1}{2}|J|^2_{\mathbb{H}^2}$ and $\nu \geq -\frac{1}{2}\log(2|\zeta|)$. Note that $e^{-2\nu}\circ\pi = |\nabla\tilde{\tau}|^2_g$ is smooth, invariant, and strictly positive at ordinary axis points, thus smooth in (ρ^2,τ). For the auxiliary radius $r = |\zeta|$, write

$$\bar{\nu}(r) := \fint_{S^2}\nu(r\omega)d\omega, \quad Q(r) := \int_1^r s^2\overline{|J|^2_{\mathbb{H}^2}}(s)ds, \quad r \geq 1.$$

Then

$$(r^2\bar{\nu}'(r))' = -\frac{1}{2}r^2\overline{|J|^2_{\mathbb{H}^2}}(r).$$

Therefore,

$$\begin{aligned}\bar\nu(1)-\bar\nu(2r) &= -\bar\nu'(1)\left(1-\frac{1}{2r}\right)+\frac{1}{2}\int_1^{2r}\frac{Q(s)}{s^2}ds\\ &\geq -|\bar\nu'(1)|+\frac{Q(r)}{4r}.\end{aligned}$$

Using $\bar\nu(2r)\geq -\frac{1}{2}\log(4r)$, we have

$$Q(r)\leq Cr(1+\log r).$$

Note that $|J|^2_{\mathbb{H}^2}$ is also subharmonic. For $|\zeta|=r\geq 2$, applying the mean-value inequality on the ball $B_{\delta_3}(\zeta,r/2)$ gives

$$\begin{aligned}|J|^2_{\mathbb{H}^2}(\zeta) &\leq Cr^{-3}\int_{B_{\delta_3}(\zeta,r/2)}|J|^2_{\mathbb{H}^2}\,dV_{\delta_3}\\ &\leq Cr^{-3}Q(2r)\leq Cr^{-2}(1+\log r).\end{aligned}$$

Next, we show that around the corner, $|J|_{\mathbb{H}^2}=\xi^{-1}+O(1)$ uniformly in direction. In the local normal coordinates (x_p,y_p) of Lemma 3.1, for $s_1=x_p^2, s_2=y_p^2, d^2=s_1+s_2$, we have

$$G_{11}=s_1A_1,\quad G_{22}=s_2A_2,\quad G_{12}=s_1s_2C,$$

where A_1,A_2,C are smooth functions of (s_1,s_2), $A_i=1+O(d^2)$, and $C=O(1)$. Their first derivatives with respect to (s_1,s_2) are bounded. Smooth torus invariance also gives

$$\begin{aligned}\rho^2 &= s_1s_2(A_1A_2-s_1s_2C^2), & A_1A_2-s_1s_2C^2 &= 1+O(d^2),\\ \tau &= (s_1-s_2)/2+E(s_1,s_2), & \xi &= d^2/2+O(d^4),\\ E &= O(d^4), & d_{(s_1,s_2)}E &= O(d^2).\end{aligned}$$

The two rows of the Jacobian of (ρ^2,τ) are

$$\big(s_2(1+O(d^2)),s_1(1+O(d^2))\big),\qquad \big(1/2+O(d^2),-1/2+O(d^2)\big).$$

Its determinant is $-d^2/2+O(d^4)$. Inverting at fixed ρ therefore gives, uniformly up to either axis,

$$\partial_\tau s_1=2s_1/d^2+O(s_1),\qquad \partial_\tau s_2=-2s_2/d^2+O(s_2).$$

Recall the hyperbolic coordinates (u,v) defined above. Then

$$u=\frac{\rho}{s_2A_2},\quad v=\frac{s_1C}{A_2}.$$

So

$$\frac{u_\tau}{u}=-\frac{\partial_\tau s_2}{s_2}-\partial_\tau\log A_2=\frac{2}{d^2}+O(1)=\frac{1}{\xi}+O(1).$$

Also $\partial_\tau v=O(s_1/d^2)$ and $\rho\asymp\sqrt{s_1s_2}$, so

$$\left|\frac{v_\tau}{u}\right|\leq C\frac{\sqrt{s_1s_2}}{s_1+s_2}\leq C.$$

Thus

$$|J|_{\mathbb{H}^2}=\frac{1}{\xi}+O(1)\text{ as }d\to 0.$$

For $\epsilon > 0$, compare the subharmonic $|J|$ with $(1+\epsilon)/\xi+\epsilon$ on an auxiliary Euclidean annulus. The inner boundary inequality holds for sufficiently small inner radius by the uniform corner estimate; the outer inequality holds for sufficiently large outer radius by decay at infinity. The ordinary axes are already included in the weak inequality and create no boundary term. Exhausting and then letting $\epsilon \downarrow 0$ gives

$$|J|_{\mathbb{H}^2} \le \frac{1}{\xi} \text{ on } \mathbb{R}^3 \setminus \{0\}.$$

So

$$\Delta_3 \log F = \xi^{-2} - |J|^2_{\mathbb{H}^2} \ge 0 \text{ on } \mathbb{R}^3 \setminus \{0\}.$$

□

3.7. **The conformal factor and a volume lower bound.**

Proposition 3.6. *The conformal factor satisfies*

$$1 \le F \le C(1+\xi), \qquad |\nabla^{\delta_3} \log F|_{\delta_3} \le C/\xi \quad (\xi > 0).$$

Moreover, $\mathrm{Vol}_g(B_g(p,R)) \ge cR^3$ *for every* $R \ge 1$.

Proof. Put $L = \log F$ and $f = \xi^{-2} - |J|^2_{\mathbb{H}^2}$. Near zero,

$$e^{-2\nu} = |\nabla^g \tilde{\tau}|^2_g = d^2 + O(d^4) = 2\xi + O(\xi^2), \qquad L = O(\xi), \qquad f = O(\xi^{-1}).$$

Set $L(0) = 0$. The punctured equation $\Delta_3 L = f$ holds distributionally on all of $\mathbb{R}^3$: test against $\chi_\epsilon \varphi$, where χ_ϵ vanishes on the ball of radius ϵ and equals one outside radius 2ϵ. The bounds $|\nabla^{\delta_3}\chi_\epsilon| \le C/\epsilon$, $|\Delta_3 \chi_\epsilon| \le C/\epsilon^2$, and $L = O(\xi)$ make the cutoff errors $O(\epsilon^2)$. The omitted source integral tends to zero as well. Hence

$$\Delta_3 L = f \quad \text{on } \mathbb{R}^3, \qquad L \ge 0, \qquad 0 \le f \le \xi^{-2}.$$

Define

$$m(r) = \frac{1}{4\pi} \int_{S^2} L(r\omega)\, d\omega, \qquad \mathcal{F}(r) = e^{m(r)}.$$

Spherical averaging, with no point source at zero, gives

$$0 \le m'(r) = \frac{1}{4\pi r^2} \int_{B_{\delta_3}(0,r)} f\, dV_{\delta_3} \le \frac{1}{r}.$$

Thus $m(r) \le \log r + C$ for $r \ge 1$. The mean-value inequality for nonnegative subharmonic L now gives the pointwise bound $L(\zeta) \le C \log(2 + |\zeta|)$.

The normalized Newton representation is

$$L(\zeta) = \frac{1}{4\pi} \int_{\mathbb{R}^3} \left(\frac{1}{|\eta|} - \frac{1}{|\zeta - \eta|} \right) f(\eta)\, d\eta. \tag{1}$$

Denote the right side by $P(\zeta)$. Splitting at $|\eta| = 2|\zeta|$ and at $|\eta| = 1$, using $f = O(|\eta|^{-1})$ near zero and $f \le |\eta|^{-2}$ elsewhere, proves absolute convergence,

$$P(\zeta) = O\big(|\zeta|(1 + |\log|\zeta||)\big) \quad (\zeta \to 0), \qquad |P(\zeta)| \le C \log(2 + |\zeta|) \quad (|\zeta| \ge 1).$$

The kernel identity gives $\Delta_3 P = f$. Hence $L - P$ is an entire harmonic function of logarithmic growth, so it is constant by the interior harmonic gradient estimate. It vanishes at zero, proving (1).

Differentiating for $\zeta \ne 0$ yields

$$|\nabla^{\delta_3} L|_{\delta_3}(\zeta) \le \frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{d\eta}{|\eta|^2 |\zeta - \eta|^2} \le \frac{C}{|\zeta|}.$$

The last integral is finite near both singularities and at infinity; scaling gives $|\zeta|^{-1}$. Integrating this gradient bound along spherical arcs and radial segments gives

$$\operatorname{osc}_{\{r/2\le|\zeta|\le r\}} L \le C, \qquad C^{-1}\mathcal{F}(r) \le F(r\omega) \le C\mathcal{F}(r).$$

Moreover $0 \le m' \le 1/r$ implies

$$\mathcal{F} \text{ is nondecreasing}, \qquad \mathcal{F}(2r) \le 2\mathcal{F}(r), \qquad 1 \le \mathcal{F}(r) \le C(1+r).$$

Consequently

$$1 \le F \le C(1+\xi).$$

Finally, we estimate the volume growth of g. Write the quotient coordinate (ρ,τ) as $\rho = \xi\sin\theta, \tau = \xi\cos\theta$. Then

$$h = \frac{F}{2\xi}(d\xi^2 + \xi^2 d\theta^2).$$

The radial segment from the corner to (r,θ) has length

$$\ell_\theta(r) = \int_0^r \sqrt{\frac{F(\xi,\theta)}{2\xi}}d\xi.$$

The bound of F implies that

$$\sqrt{2r} \le \ell_\theta(r) \le C(\sqrt{r}+r).$$

Since quotient distance from $\pi(p)$ equals g-distance from p, for any q over (r,θ), we have

$$d_g(p,q) \le \ell_\theta(r).$$

Since $\rho = \sqrt{\det G}$, we have

$$\pi_*(dV_g) = 4\pi^2\rho dV_h = 2\pi^2 F(\xi,\theta)\xi\sin\theta d\xi d\theta.$$

Fix a small constant $\epsilon > 0$. For all large radius R, for all θ, we have

$$\ell_\theta(\epsilon R) \le \frac{1}{2}R, \quad \ell_\theta(R'_\theta) = R,$$

for some $\epsilon R < R'_\theta$. So the domain over the radial region $\xi < R'_\theta$ is inside $B_g(p,R)$. Note that

$$\frac{1}{2}R \le \ell_\theta(R'_\theta) - \ell_\theta(\epsilon R) = \int_{\epsilon R}^{R'_\theta}\sqrt{\frac{F(\xi,\theta)}{2\xi}}d\xi \le \left(\frac{1}{2\epsilon R}\int_{\epsilon R}^{R'_\theta}\xi F(\xi,\theta)d\xi\right)^{\frac{1}{2}}.$$

Therefore,

$$\begin{aligned}\operatorname{Vol}_g(B_g(p,R)) &\ge 2\pi^2\int_0^\pi \sin\theta\int_{\epsilon R}^{R'_\theta}\xi F(\xi,\theta)d\xi d\theta \\ &\ge 2\pi^2\epsilon R^3.\end{aligned}$$

Reducing c to cover bounded radii gives

$$\operatorname{Vol}_g(B_g(p,R)) \ge cR^3, \quad \forall R \ge 1.$$

□

3.8. **The asymptotic dichotomy.**

Proposition 3.7. *Either g is Euclidean, or, uniformly as $\xi \to \infty$,*

$$\xi |J|_{\mathbb{H}^2} \to 0, \qquad \xi |\nabla^{\delta_3} \nu|_{\delta_3} \to 0, \qquad F = 2\xi e^{2\nu} \to \infty.$$

In the latter case, after target isometries, every subsequential interior limit of $H(t\rho, t\tau)$ is a unit-speed hyperbolic geodesic parametrized by $\log \rho$.

Proof. Set $L = \log F$, $a = |J|_{\mathbb{H}^2}$ and $q(r) = r\bar{a}(r)$, where the bar denotes the spherical average in $\mathbb{R}^3$. Since $\xi = |\zeta|$, $a(\zeta) \le \frac{1}{|\zeta|}$ and $a(\zeta) = \frac{1}{|\zeta|} + O(1)$ as $\zeta \to 0$, so $0 \le q \le 1$ and $q(0+) = 1$. For a nonnegative $\varphi \in C_c^\infty((0,\infty))$,

$$\Delta_3 \left(\frac{\varphi(r)}{r} \right) = \frac{\varphi''(r)}{r}, \qquad 4\pi \int_0^\infty q(r)\varphi''(r)\, dr = \int_{\mathbb{R}^3} r^{-1} \varphi(\Delta_3 a) dV_{\delta_3} \ge 0.$$

Thus q is convex. A convex function bounded above on $(0,\infty)$ is nonincreasing, so $\lim_{r\to\infty} q(r) = c$ for some $c \in [0,1]$.

We first prove that, for every $1 \le p < \infty$, as $t \to \infty$,

$$a_t(\zeta) := t a(t\zeta) \to c/|\zeta| \quad \text{in } L^p_{\mathrm{loc}}(\mathbb{R}^3 \setminus \{0\}). \tag{2}$$

Indeed, a_t is nonnegative, subharmonic, locally Lipschitz on $\mathbb{R}^3 \setminus \{0\}$, and bounded by $|\zeta|^{-1}$. For a cutoff χ supported away from zero, integration by parts gives

$$\int \chi^2 |\nabla^{\delta_3} a_t|^2_{\delta_3} \le 4 \int a_t^2 |\nabla^{\delta_3} \chi|^2_{\delta_3}.$$

Rellich compactness therefore, up to a subsequence $t_j \to \infty$, $a_{t_j} \to a_\infty$ in strong local L^2-topology, with $0 \le a_\infty \le |\zeta|^{-1}$ and $\Delta_3 a_\infty \ge 0$. Since $r\overline{a_{t_j}}(r) = q(t_j r) \to c$ as $t_j \to \infty$, by Jensen's inequality, we know $r\overline{a_\infty}(r) = c$ for almost every r. Together with mean value inequality of $\Delta_3 a_\infty \ge 0$, this proves the mean value equality, thus $\Delta_3 a_\infty = 0$. The Kelvin transform

$$\widehat{a_\infty}(\zeta) = |\zeta|^{-1} a_\infty(\zeta/|\zeta|^2)$$

is harmonic and bounded between zero and one on $\mathbb{R}^3 \setminus \{0\}$, and thus is a constant by the Liouville theorem. Using the spherical mean identifies again, we obtain $a_\infty = c/|\zeta|$, independent of the choice of subsequence. The local uniform bound upgrades convergence to every finite L^p. It also gives $a_t^2 \to c^2/|\zeta|^2$ in every finite local L^p.

Fix $\zeta_0 = (1,0,0)$, so $(\rho,\tau) = (1,0)$, and set $L_t(\zeta) = L(t\zeta) - L(t\zeta_0) = \log \frac{|\zeta|}{|\zeta_0|} + 2\nu(t\zeta) - 2\nu(t\zeta_0)$. The gradient estimate and

$$\Delta_3 L_t = |\zeta|^{-2} - a_t^2$$

give local $W^{2,p}$ bounds on every annulus, including its rotation axis. Subsequential limits satisfy

$$\Delta_3 L_\infty = (1-c^2)|\zeta|^{-2}, \qquad |\nabla^{\delta_3} L_\infty|_{\delta_3} \le C/|\zeta|, \qquad L_\infty(\zeta_0) = 0.$$

The harmonic function $L_\infty - (1-c^2)\log|\zeta|$ has at most logarithmic growth both at zero and infinity. So it can be extended to a global harmonic function and thus a constant zero by Liouville theorem. Thus, for every $0 < \alpha < 1$,

$$L_t \to (1-c^2)\log|\zeta|, \quad \nu(t\zeta) - \nu(t\zeta_0) \to -\tfrac{1}{2}c^2 \log|\zeta| \ \text{ in } C^{1,\alpha}_{\mathrm{loc}}(\mathbb{R}^3 \setminus \{0\}). \tag{3}$$

Next, we show that $0 < c < 1$ is impossible. For each $t > 0$, let I_t be a target isometry that sends $H(t\zeta_0)$ to $(1,0) \in \mathbb{H}^2$. More precisely, if $H(t\zeta_0) = (U_t, V_t)$, then we can take $I_t(u,v) = \left(\frac{u}{U_t}, \frac{v-V_t}{U_t}\right)$. Define $\mathcal{H}_t(\zeta) = I_t(H(t\zeta))$. Then

$$\begin{aligned}
|\partial_\tau \mathcal{H}_t|^2_{\mathbb{H}^2}(\zeta) &= t^2 |H_\tau|^2_{\mathbb{H}^2}(t\zeta) = a_t^2(\zeta),\\
|\partial_\rho \mathcal{H}_t|^2_{\mathbb{H}^2}(\zeta) &= t^2 |H_\rho(t\rho, t\tau)|^2_{\mathbb{H}^2} = \rho^{-2} + 4\rho^{-1}\partial_\rho \nu^t(\zeta) + a_t^2(\zeta),\\
\langle \partial_\rho \mathcal{H}_t, \partial_\tau \mathcal{H}_t\rangle_{\mathbb{H}^2}(\zeta) &= t^2 \langle H_\rho, H_\tau\rangle_{\mathbb{H}^2}(t\rho, t\tau) = 2t\rho^{-1}\nu_\tau(t\rho, t\tau) = 2\rho^{-1}\partial_\tau \nu^t(\zeta),
\end{aligned}$$

where $\nu^t(\zeta) := \nu(t\zeta) - \nu(t\zeta_0)$. Interior elliptic estimates for the harmonic-map equation then yield smooth subsequential convergence $\mathcal{H}_{t_j}(\zeta) \to \mathcal{H}_\infty(\zeta)$ on $\rho > 0$. By (2)–(3),

$$|\partial_\tau \mathcal{H}_\infty|^2_{\mathbb{H}^2} = \frac{c^2}{|\zeta|^2}, \qquad |\partial_\rho \mathcal{H}_\infty|^2_{\mathbb{H}^2} = \frac{1}{\rho^2} - \frac{c^2}{|\zeta|^2}, \qquad \langle \partial_\rho \mathcal{H}_\infty, \partial_\tau \mathcal{H}_\infty\rangle_{\mathbb{H}^2} = -\frac{c^2\tau}{\rho|\zeta|^2}.$$

Introduce only for this computation $\sigma = \log\rho$ and $\upsilon = \operatorname{arsinh}(\tau/\rho)$. Substitution gives

$$\mathcal{H}_\infty^* g_{\mathbb{H}^2} = (1-c^2)d\sigma^2 + c^2 d\upsilon^2 \text{ on } X^\circ.$$

For $0 < c < 1$ this is a positive definite flat metric, so $\mathcal{H}_\infty$ is a local diffeomorphism whose pullback metric has curvature zero, a contradiction with the target curvature -1. Thus $c = 0$ or 1.

If $c = 1$, monotonicity and $q(0+) = 1$ give $q \equiv 1$. The pointwise inequality $a \le \xi^{-1}$ then gives equality everywhere. Hence $\Delta_3 L = 0$ on all of $\mathbb{R}^3$; $L \ge 0$ and $L(0) = 0$ imply $L = 0$. Thus $F \equiv 1$, and the equality case of Colding's gradient estimate gives the Euclidean metric [9, Theorem 3.1].

In the nonflat case $c = 0$, (2) gives

$$a_t \to 0 \text{ in } L^2_{\mathrm{loc}}(\mathbb{R}^3 \setminus \{0\}).$$

The mean-value inequality for the subharmonic a_t implies that $a_t(\zeta) = ta(t\zeta) \to 0$ locally uniformly on $\mathbb{R}^3 \setminus \{0\}$. In particular, $\xi a \to 0$ uniformly as $\xi \to \infty$. Equation (3) gives $\nu^t(\zeta) \to 0$ in $C^{1,\alpha}_{\mathrm{loc}}(\mathbb{R}^3 \setminus \{0\})$, so $|\nabla^{\delta_3}\nu^t|(\zeta) = t|\nabla^{\delta_3}\nu|(t\zeta) \to 0$ locally uniformly. In particular, $\xi|\nabla^{\delta_3}\nu|_{\delta_3} \to 0$ as $\xi \to \infty$. Integration along rays gives $\nu = o(\log\xi)$ uniformly. Therefore $\log F = \log 2\xi + 2\nu = \log 2\xi + o(\log\xi) \to \infty$ uniformly as $\xi \to \infty$. Finally the interior limit has $\partial_\tau \mathcal{H}_\infty = 0$ and $|\partial_\rho \mathcal{H}_\infty|_{\mathbb{H}^2} = 1/\rho$; its harmonic-map equation becomes the unit-speed geodesic equation in $\sigma = \log\rho$. □

We give a consequence of this asymptotic behavior. Using $\xi|\nabla^{\delta_2}\nu|_{\delta_2} \to 0$ uniformly as $\xi \to \infty$, we know $\sup_{\xi=s}|\nu(\rho,\tau) - \nu(s,0)| = o(1)$ as $s \to \infty$. Thus $e^{\nu(\rho,\tau)} = (1+o(1))e^{\nu(s,0)}$ uniformly, and

$$d_g(p,q) = (1+o(1))r_s \text{ uniformly for } \tilde{\xi}(q) = s, \quad \text{as } s \to \infty,$$

where $r_s := se^{\nu(s,0)}$.

3.9. **Adapted Killing fields and uniform estimates.** Assume g is nonflat. We choose a constant torus Killing field on each large annulus and estimate its logarithmic derivative through both axes. Recall $r_s = se^{\nu(s,0)}$.

Lemma 3.8. *For every sufficiently large s, there is a constant ω^s with $1/2 \le |\omega^s| \le 2$ such that the globally defined Killing field*

$$K_s = -\omega^s X_1 + X_2$$

satisfies

$$\frac{|\nabla^g K_s|_g}{|K_s|_g} \le \frac{C}{r_s} \quad \text{on } \{s \le \tilde{\xi} \le 2s\}.$$

Proof. We first control the collapsing circle near each axis, then use a scalar elliptic equation to choose one constant coefficient vector for both axes. Set

$$\sigma_s = s \sup_{s/16 \le \xi \le 16s,\ \rho>0} \left(|J|_{\mathbb{H}^2} + |\nabla^{\delta_3} \nu|_{\delta_3} \right) \to 0.$$

Path integration on this larger annulus gives

$$e^{2\nu(\rho,\tau)} = e^{2\nu(s,0)}(1 + O(\sigma_s)). \tag{4}$$

The negative axis $\tau < 0$ is where X_1 collapses.

Consider the region $0 < \rho < 3s$, $-4s < \tau < -s/4$, and put $k = G_{11}/\rho^2$. Recall that in Lemma 3.2, for q over $(0, \tau)$, we have $|X_1|(q) = 0$,

$$|\nabla \tilde{\tau}|_g^2(q) = |X_2|_g^2(q) \neq 0.$$

On the quotient, we obtain $|d\tau|_h^2(0,\tau) = e^{-2\nu(0,\tau)} = G_{22}(0,\tau)$. On the other hand, $k^{-1} = G_{22} - \frac{G_{12}^2}{G_{11}}$, so $k^{-1}(0,\tau) = G_{22}(0,\tau)$. Thus

$$k(0,\tau) = e^{2\nu(0,\tau)},$$

and k is smooth in (ρ^2, τ). The Weyl equation gives

$$\left| |H_\rho|_{\mathbb{H}^2}^2 - \rho^{-2} \right| \le \frac{C\sigma_s}{s\rho}, \qquad |dH|_{\delta_3, \mathbb{H}^2} \le C/\rho.$$

Recall that $\Omega = \mathbb{R}^3 \setminus \mathcal{A}$ is the auxiliary Euclidean space. Fix $\zeta_0 \in \Omega$ with $s/8 \le \xi(\zeta_0) \le 8s$, and set $\rho_0 = \rho(\zeta_0)$ and $R = \rho_0/8$. Then $B_{\delta_3}(\zeta_0, 2R)$ avoids the axis, and ρ is comparable to ρ_0 throughout this ball. Similarly as in previous section, we normalize the target by an isometry. More precisely, if $H(\zeta_0) = (u_0, v_0)$, then take $I(u,v) = \left(\frac{u}{u_0}, \frac{v - v_0}{u_0} \right)$ so that $I(H(\zeta_0)) = (1,0) \in \mathbb{H}^2$. Define, for $y \in B_2(0) \subset \mathbb{R}^3$,

$$\tilde{H}(y) = I(H(\zeta_0 + Ry)).$$

Then $\tilde{H}(0) = (1,0)$ and the gradient estimate gives

$$|d\tilde{H}|_{\delta_3, \mathbb{H}^2} \le C.$$

The harmonic-map equation gives uniform interior bounds for the derivatives of $\tilde{H}$, by elliptic bootstrapping. Note that $\tilde{J}(y) = dI(J(\zeta_0 + Ry))$ satisfies the Jacobi equation, whose coefficients are controlled by the bounds of $\tilde{H}, d\tilde{H}$. So interior estimates give

$$|\tilde{D}\tilde{J}|_{\delta_3, \mathbb{H}^2}(0) \le C \sup_{B_1(0)} |\tilde{J}|_{\mathbb{H}^2}.$$

Therefore,

$$|D_\tau J|_{\mathbb{H}^2}(\rho,\tau) \le \frac{C}{\rho} \sup |J|_{\mathbb{H}^2} \le \frac{C\sigma_s}{s\rho} \quad \text{on } \{s/8 \le \xi \le 8s\}. \tag{5}$$

Use the upper-half-plane chart $u(H) = H_{11}^{-1}$, $v(H) = H_{12}/H_{11}$, obtained from the earlier chart by interchanging the two matrix indices. Then $H_{11} = \rho k = 1/u(H)$. Note that $\mathrm{Hess}_{\mathbb{H}^2} \frac{1}{u} = \frac{1}{u} g_{\mathbb{H}^2}$ and $|d\frac{1}{u}|_{\mathbb{H}^2} = \frac{1}{u}$. For simplicity, set $\mathcal{T} = \frac{1}{u}$ and $T = H_{11} = \mathcal{T}(H)$. The harmonic-map equation yields

$$T_{\rho\rho} + \rho^{-1} T_\rho = |H_\rho|_{\mathbb{H}^2}^2 T - (d\mathcal{T})_H (D_\tau J).$$

Substitution of $T = \rho k$ and (5) gives

$$\left|\partial_\rho(\rho^3 k_\rho)\right| \le \frac{C\sigma_s}{s}\rho^2 k.$$

Fix τ. Since $\rho^3 k_\rho \to 0$ at the axis, taking integrations give

$$|k_\rho(\rho,\tau)| \le \frac{C\sigma_s}{s}\sup_{0\le r\le\rho} k(r,\tau), \qquad |k(\rho,\tau)-k(0,\tau)| \le \frac{C\sigma_s\rho}{s}\sup_{0\le r\le\rho} k(r,\tau).$$

For $\rho \le 3s$, absorb the last supremum term when s is large to obtain

$$|\partial_\rho \log k(\rho,\tau)| \le \frac{C\sigma_s}{s}, \qquad |k(\rho,\tau)-k(0,\tau)| \le \frac{C\sigma_s\rho}{s}k(0,\tau).$$

Also

$$|\partial_\tau \log k| = |\partial_\tau \log T| = \frac{|(H_{11})_\tau|}{|H_{11}|} \le |J|_{\mathbb{H}^2}.$$

Together with (4) and $k(0,\tau) = e^{2\nu(0,\tau)}$, these facts give

$$k = e^{2\nu(s,0)}(1+O(\sigma_s)), \quad |\nabla^{\delta_2}\log k|_{\delta_2} \le C\sigma_s/s, \tag{6}$$

where $0 \le \rho \le 3s$, $-4s \le \tau \le -s/4$, for all uniformly large s.

We now write

$$G_{ij}d\theta^i d\theta^j = k\rho^2(d\theta^1+\omega d\theta^2)^2 + k^{-1}(d\theta^2)^2, \qquad \omega = G_{12}/G_{11}. \tag{7}$$

In Lemma 3.2, we have shown that both k and ω are smooth in (ρ^2,τ), aross $\rho = 0$. In the target hyperbolic chart (u,v), $U := u(H) = (k\rho)^{-1}$, $V := v(H) = \omega$. The V-component of the harmonic map equation gives $\Delta_3 V - 2U^{-1}\langle dU, dV\rangle_{\delta_3} = 0$. After substituting $U = (k\rho)^{-1}$ and $V = \omega$, the equation becomes

$$\partial_\rho^2\omega + \frac{3}{\rho}\partial_\rho\omega + \partial_\tau^2\omega + 2\left(\frac{k_\rho}{k}\partial_\rho\omega + \frac{k_\tau}{k}\partial_\tau\omega\right) = 0.$$

To simplify the arguments, we introduce the auxiliary Euclidean space $(\mathbb{R}^5,\delta_5)$ with coordinates $(z,\tau)\in\mathbb{R}^4\times\mathbb{R}$ and identify $\rho = |z|$ so that $\Delta_5 = \partial_\rho^2 + 3\rho^{-1}\partial_\rho + \partial_\tau^2$. Then we can rewrite above equation as

$$\mathrm{div}_{\delta_5}(k^2\nabla^{\delta_5}\omega) = 0. \tag{8}$$

Smoothness in ρ^2 makes this equation regular across $z = 0$.

The target hyperbolic metric gives

$$|H_\rho|^2_{\mathbb{H}^2} = (\rho^{-1}+k_\rho/k)^2 + (k\rho)^2(\partial_\rho\omega)^2, \qquad |H_\tau|^2_{\mathbb{H}^2} = (k_\tau/k)^2 + (k\rho)^2(\partial_\tau\omega)^2.$$

Using the Weyl equation gives

$$(k\rho)^2(\partial_\rho\omega)^2 = \frac{4\nu_\rho - 2\partial_\rho\log k}{\rho} + |J|^2_{\mathbb{H}^2} - (\partial_\rho\log k)^2.$$

Using (6) and only $\sigma_s \le 1$, we obtain

$$|\nabla^{\delta_5}\omega|^2_{\delta_5} \le Ce^{-4\nu(s,0)}\left(\frac{1}{s\rho^3}+\frac{1}{s^2\rho^2}\right).$$

On the cylinder $\mathcal{C}_s = \{|z| < 3s,\ -4s < \tau < -s/4\}$, integration with $dV_{\delta_5} = |\mathbb{S}^3|\rho^3 d\rho\, d\tau$ therefore gives

$$\int_{\mathcal{C}_s}|\nabla^{\delta_5}\omega|^2_{\delta_5}dV_{\delta_5} \le Cse^{-4\nu(s,0)} = Cs^5/r_s^4.$$

By (6), the coefficient $e^{-4\nu(s,0)}k^2$ is bounded above and below by positive constants, and its δ_5-gradient is bounded by C/s. We apply the interior gradient estimate to (8) after multiplying by the constant $e^{-4\nu(s,0)}$. Consider the smaller band and its lift

$$\mathcal{B}_s^- = \{0 \le \rho \le 2s,\ -3s \le \tau \le -s/2\}, \qquad \widehat{\mathcal{B}}_s^- = \{|z| \le 2s,\ -3s \le \tau \le -s/2\}.$$

Every point $q \in \widehat{\mathcal{B}}_s^-$ has Euclidean distance at least $s/4$ from $\partial\mathcal{C}_s$, so the Euclidean ball $B_{s/8}(q)$ lies in $\mathcal{C}_s$. Then the interior scalar gradient estimate and Poincaré's inequality give

$$|\nabla^{\delta_5}\omega|_{\delta_5}(q) \le \frac{C}{s}\left(\fint_{B_{s/8}(q)} |\omega - \bar{\omega}|^2 dV_{\delta_5}\right)^{1/2} \le C\left(\fint_{B_{s/8}(q)} |\nabla^{\delta_5}\omega|_{\delta_5}^2 dV_{\delta_5}\right)^{1/2}.$$

Thus

$$\sup_{\widehat{\mathcal{B}}_s^-} |\nabla^{\delta_5}\omega|_{\delta_5} \le C\left(s^{-5}\int_{\mathcal{C}_s} |\nabla^{\delta_5}\omega|_{\delta_5}^2 dV_{\delta_5}\right)^{1/2} \le Cr_s^{-2}, \tag{9}$$

which implies that

$$\sup_{\mathcal{B}_s^-} |d\omega|_{\delta_2} \le Cr_s^{-2}. \tag{10}$$

On the reflected band $\mathcal{B}_s^+ = \{0 \le \rho \le 2s,\ s/2 \le \tau \le 3s\}$, interchange X_1, X_2 and use $\widetilde{k} = G_{22}/\rho^2$, $\widetilde{\omega} = G_{12}/G_{22}$.

Set $\omega^s = \omega(s,0) = G_{12}(s,0)/G_{11}(s,0)$. The bound for k at $(s,-s)$ controls G_{11}, and its positive-axis counterpart at (s,s) controls G_{22}. Since

$$|\partial_\tau \log G_{ii}| \le |J|_{\mathbb{H}^2} \le \sigma_s/s \quad \text{on } \rho = s,\ -s \le \tau \le s,$$

integration gives, on this whole segment,

$$G_{11} = r_s^2(1 + O(\sigma_s)), \qquad G_{22} = r_s^2(1 + O(\sigma_s)).$$

At $(s,0)$, $\det G = s^2$, and $s/r_s^2 = 2/F(s,0) \to 0$ as $s \to \infty$. Hence $(\omega^s)^2 = G_{22}/G_{11} - s^2/G_{11}^2 \to 1$, so $1/2 \le |\omega^s| \le 2$ for large s. The estimate of $|J|_{\mathbb{H}^2}$ also gives

$$|\partial_\tau \omega| \le C/r_s^2 \quad \text{on } \rho = s,\ |\tau| \le s.$$

Combining this with (10) yields

$$|\omega - \omega^s| \le Cs/r_s^2 \quad \text{on } \mathcal{B}_s^-. \tag{11}$$

For $\mathcal{B}_s^+$, use the exact identity

$$(\omega^s)^{-1} - \widetilde{\omega}(s,0) = \frac{s^2}{G_{12}(s,0)G_{22}(s,0)},$$

whose absolute value is at most $Cs^2/r_s^4 \le Cs/r_s^2$. The analogous segment and twist estimates therefore give

$$|\widetilde{\omega} - (\omega^s)^{-1}| \le Cs/r_s^2 \quad \text{on } \mathcal{B}_s^+. \tag{12}$$

Now we can construct the adapted Killing fields. For a constant vector $\mathbf{v} = (-c_0, 1)^T$, set $K = -c_0X_1 + X_2 = (X_1, X_2)\mathbf{v}$. Since

$$d\theta^1(K) = -c_0, d\theta^2(K) = 1, (d\theta^1 + \omega d\theta^2)(K) = \omega - c_0,$$

by (7),

$$|K|_g^2 = k\rho^2(\omega - c_0)^2 + k^{-1}.$$

Also,

$$K^{\flat_g} = (G\mathbf{v})_i d\theta^i = k\rho^2(\omega - c_0)(d\theta^1 + \omega d\theta^2) + k^{-1}d\theta^2.$$

So

$$\begin{aligned} dK^{\flat_g} &= d\left(k\rho^2(\omega - c_0)\right) \wedge (d\theta^1 + \omega d\theta^2) + k\rho^2(\omega - c_0)d\omega \wedge d\theta^2 - k^{-2}dk \wedge d\theta^2 \\ &= \left(\rho^2(\omega - c_0)dk + 2k\rho(\omega - c_0)d\rho + k\rho^2 d\omega\right) \wedge (d\theta^1 + \omega d\theta^2) \\ &\quad + \left(k\rho^2(\omega - c_0)d\omega - k^{-2}dk\right) \wedge d\theta^2. \end{aligned}$$

Since K is Killing,

$$\begin{aligned} 2|\nabla^g K|_g^2 &= |dK^{\flat_g}|_g^2 \\ &= \frac{1}{k\rho^2}\left|\left(\rho^2(\omega - c_0)dk + 2k\rho(\omega - c_0)d\rho + k\rho^2 d\omega\right)\right|_h^2 \\ &\quad + k\left|\left(k\rho^2(\omega - c_0)d\omega - k^{-2}dk\right)\right|_h^2. \end{aligned}$$

Consequently,

$$\begin{aligned} |\nabla^g K|_g^2 &\le Ce^{-2\nu}\left(k\rho^2(\omega - c_0)^2|d\log k|_{\delta_2}^2 + k(\omega - c_0)^2|d\rho|_{\delta_2}^2 + k\rho^2|d\omega|_{\delta_2}^2\right) \\ &\quad + Ce^{-2\nu}\left(k^3\rho^4(\omega - c_0)^2|d\omega|_{\delta_2}^2 + k^{-1}|d\log k|_{\delta_2}^2\right) \\ &\le Ce^{-2\nu}\left(|K|_g^2|d\log k|_{\delta_2}^2 + k^2\rho^2|K|_g^2|d\omega|_{\delta_2}^2 + k(\omega - c_0)^2\right). \end{aligned}$$

Thus

$$\frac{|\nabla^g K|_g^2}{|K|_g^2} \le Ce^{-2\nu}\left(|d\log k|_{\delta_2}^2 + k^2\rho^2|d\omega|_{\delta_2}^2 + k^2(\omega - c_0)^2\right). \tag{13}$$

On $\mathcal{B}_s^-$ take $c_0 = \omega^s$. Since $k \le Cr_s^2/s^2$, (6), (10), and (11) bound each term in parentheses by C/s^2. Thus (4) gives $|\nabla^g K_s|_g/|K_s|_g \le C/r_s$. On $\mathcal{B}_s^+$ write $K_s = -\omega^s(X_1 - (\omega^s)^{-1}X_2)$ and use (12) in the same way. These two bands cover the part of $s \le \xi \le 2s$ and $\rho < s/2$.

It remains to control the part $s \le \xi \le 2s$ and $\rho \ge s/2$. Note that

$$dK^{\flat_g} = \sum_{\alpha=\rho,\tau}\sum_i (\partial_\alpha G\mathbf{v})_i\, dx^\alpha \wedge d\theta^i,$$

so

$$\begin{aligned} |\nabla^g K|_g^2 &= \frac{1}{2}e^{-2\nu}\sum_{\alpha=\rho,\tau}(\partial_\alpha G\mathbf{v})^T G^{-1}(\partial_\alpha G\mathbf{v}) \\ &\le \frac{1}{2}e^{-2\nu}\sum_\alpha \operatorname{tr}(G^{-1}\partial_\alpha G G^{-1}\partial_\alpha G)\mathbf{v}^T G\mathbf{v} \\ &= \frac{1}{2}e^{-2\nu}\sum_\alpha \left(2\rho^{-2}(\partial_\alpha\rho)^2 + 2|H_\alpha|_{\mathbb{H}^2}^2\right)|K|_g^2 \\ &\le e^{-2\nu}(\rho^{-2} + |dH|_{\delta_3,\mathbb{H}^2}^2)|K|_g^2. \end{aligned}$$

The Weyl equation gives $|dH|_{\delta_3,\mathbb{H}^2} \le C/\rho$ when $s \le \xi \le 2s$. If $\rho \ge s/2$, we also have $|\nabla^g K_s|_g/|K_s|_g \le Ce^{-\nu}/s \le C/r_s$. This proves the stated estimate on all of $\{s \le \tilde{\xi} \le 2s\}$. □

3.10. **Curvature energy and the classification.** We apply the local Euler and Pontryagin transgression formulas for a Killing field [20, Sections 2.1–2.2]. The Euler identity first establishes finite curvature energy; the Pontryagin identity then forces one Weyl half to vanish.

Let $\mathcal{A}_s = \{s \le \tilde{\xi} \le 2s\} \subset M$, and recall $r_s = se^{\nu(s,0)}$. So

$$\begin{aligned} \mathrm{Vol}_g(\mathcal{A}_s) &= 4\pi^2 \int_{\substack{s\le\xi\le 2s\\ \rho>0}} e^{2\nu}\rho\, d\rho\, d\tau \le Ce^{2\nu(s,0)}s^3 \le CF(s,0)s^2, \\ v_s &:= r_s^{-4}\mathrm{Vol}_g(\mathcal{A}_s) \le \frac{C}{F(s,0)} \to 0. \end{aligned} \tag{14}$$

For a Ricci-flat four-metric the Euler form is

$$e_g = \frac{1}{8\pi^2}(|W^+|^2_{g,\mathrm{op}} + |W^-|^2_{g,\mathrm{op}})\, dV_g \ge 0.$$

Here the norms are Hilbert–Schmidt norms of the curvature operators on $\Lambda^{\pm}_g$; the four-index tensor norm satisfies $|\mathrm{Rm}_g|^2_g = 4(|W^+|^2_{g,\mathrm{op}} + |W^-|^2_{g,\mathrm{op}})$.

Proposition 3.9. *Assume g is not flat. Then*

$$\int_M e_g = 1, \qquad \int_M |\mathrm{Rm}_g|^2_g\, dV_g = 32\pi^2 < \infty.$$

Proof. Let $K_s = -\omega^s X_1 + X_2$ be the Killing field in previous lemma. Both coefficients of K_s are nonzero, so its only zero is p. Its linearization there is a pair of nonzero planar rotations and has positive determinant, hence index one. More precisely, using the torus action splitting of T_pM, $(\nabla^g K_s)_p : T_pM \to T_pM$ has the form

$$\begin{pmatrix} -\omega^s J & 0 \\ 0 & J \end{pmatrix}, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

so $\det(\nabla^g K_s)_p = (\omega^s)^2 > 0$. We use Chern's classical Euler transgression [7, Sections 2–3], [8, Section 1, (9)–(11), and Section 2]. Let $\varpi : S(TM) \to M$ be the unit tangent sphere bundle. Chern's angular three-form Ψ satisfies

$$d\Psi = -\varpi^* e_g, \qquad \int_{S(T_xM)} \Psi = 1,$$

where each fiber has its boundary orientation as the unit sphere in T_xM. The unit vector field $u_s = K_s/|K_s|_g$ is a section of this bundle on $M \setminus \{p\}$. Therefore the three-form

$$T^e_s := -u_s^*\Psi \quad \text{satisfies} \quad dT^e_s = e_g \quad \text{on } M \setminus \{p\}.$$

In dimension four, Chern's formula contains terms cubic in $\nabla^g u_s$ and terms linear in both $\nabla^g u_s$ and curvature. Since $|\nabla^g u_s|_g \le |\nabla^g K_s|_g/|K_s|_g$, it gives

$$|T^e_s|_g \le C(\ell_s|\mathrm{Rm}_g|_g + \ell_s^3), \qquad \ell_s := |\nabla^g K_s|_g/|K_s|_g.$$

Nondegeneracy of the zero gives $\ell_s = O(r^{-1})$ for $r = d_g(p,\cdot)$, so T^e_s is locally integrable at p. The fiber normalization and the local index give

$$\lim_{\epsilon\downarrow 0} \int_{\partial B_g(p,\epsilon)} T^e_s = -\operatorname{ind}_p(K_s) = -1,$$

with the sphere oriented outward from p. Stokes's theorem on the complement of this ball consequently yields

$$dT^e_s = e_g - \delta_p \quad \text{distributionally on } M,$$

where δ_p is the Dirac four-current, $\langle \delta_p, \varphi \rangle = \varphi(p)$.

Choose $\eta_s = \eta(\xi/s)$, with $\eta = 1$ on $[0,1]$, $\eta = 0$ on $[2,\infty)$ and $0 \le \eta \le 1$. On $\mathcal{A}_s$,

$$|d\eta_s| = s^{-1}|\eta'| \cdot |d\xi|_g \le Cs^{-1}e^{-\nu} \le Cs^{-1}e^{-\nu(s,0)} = Cr_s^{-1},$$

so $|d\eta_s|_g + \ell_s \le Cr_s^{-1}$. Set $I_s = \int_M \eta_s^2 e_g$. Since $\eta_s(p) = 1$,

$$I_s - 1 = -\int_M d\eta_s^2 \wedge T_s^e = -2\int_{\mathcal{A}_s} \eta_s d\eta_s \wedge T_s^e.$$

Cauchy–Schwarz and (14) give

$$\begin{aligned} |I_s - 1| &\le Cr_s^{-2} \int_{\mathcal{A}_s} \eta_s |\mathrm{Rm}_g|_g \, dV_g + Cv_s \\ &\le C\sqrt{v_s I_s} + Cv_s. \end{aligned}$$

As $s \to \infty$, Young's inequality first bounds I_s, and the same estimate then gives $I_s \to 1$. Properness of ξ gives $\eta_s \to 1$ pointwise. Fatou's lemma establishes finite total Euler energy, after which dominated convergence shows that its value is one. □

Define the two nonnegative chiral energies

$$\mathcal{E}_\pm = \frac{1}{8\pi^2}\int_M |W^\pm|^2_{g,\mathrm{op}} dV_g, \qquad \mathcal{E}_+ + \mathcal{E}_- = 1.$$

For the curvature two-form Ω^g of ∇^g, our first Pontryagin form is

$$p_1(\nabla^g) = -\frac{1}{8\pi^2}\mathrm{tr}(\Omega^g \wedge \Omega^g), \qquad P := \int_M p_1 = 2(\mathcal{E}_+ - \mathcal{E}_-).$$

In particular, $|P| \le 2$, with equality exactly when one Weyl half vanishes.

Proof of Theorem 1.2. It remains to consider the nonflat case. As in the Euler argument, we transgress the curvature form away from the zero and compute its local residue. For the Pontryagin form we use the Killing-field construction in [20, Section 2.1, (3), p. 3119]. Let $K = a_1X_1 + a_2X_2$, where $a_1a_2 \neq 0$, and on $M \setminus \{p\}$ set

$$\alpha = \frac{K^{\flat_g}}{|K|_g^2}, \qquad B(V) = \nabla^g_V K.$$

Here B is a skew-adjoint endomorphism field. The transgression three-form is

$$T_K = -\frac{1}{4\pi^2}\alpha \wedge \mathrm{tr}(B\Omega^g) + \frac{1}{8\pi^2}\mathrm{tr}(B^2)\alpha \wedge d\alpha, \tag{15}$$

where the trace is over tangent-space endomorphisms. To see why this form exists, compare ∇^g with the metric connection

$$\nabla^0_V Y = \nabla^g_V Y - \alpha(V)B(Y).$$

Choose locally an orthonormal frame invariant under the flow of K. If ϑ is the connection matrix of ∇^g in this frame, then $\vartheta(K) = B$, because $[K, e_i] = 0$ implies $\nabla^g_K e_i = \nabla^g_{e_i} K$. Hence the connection matrix $\vartheta^0 = \vartheta - \alpha B$ is invariant and annihilates K. Cartan's formula gives $\iota_K \Omega^0 = 0$ for its curvature Ω^0. Consequently $\mathrm{tr}(\Omega^0 \wedge \Omega^0) = 0$ in dimension four. The standard transgression between these two connections gives (15) and $dT_K = p_1(\nabla^g)$ on $M \setminus \{p\}$. Moreover,

$$|\alpha|_g = |K|_g^{-1}, \qquad |d\alpha|_g \le C\frac{|\nabla^g K|_g}{|K|_g^2},$$

so (15) gives

$$|T_K|_g \le C(\ell_K |\mathrm{Rm}_g|_g + \ell_K^3), \qquad \ell_K = \frac{|\nabla^g K|_g}{|K|_g}.$$

Use the orientation of the ordered primitive normal-coordinate planes at p, with signed rotation speeds a_1, a_2. In their tangent model $(\mathbb{C}^2, \delta_4)$, where δ_4 is Euclidean, write $K_0(z_1, z_2) = (ia_1 z_1, ia_2 z_2)$ and $\alpha_0 = K_0^{\flat_{\delta_4}}/|K_0|^2_{\delta_4}$. The linearization $B_0 = \mathrm{diag}(a_1 J, a_2 J)$ satisfies $\mathrm{tr}(B_0^2) = -2(a_1^2 + a_2^2)$. On the unit sphere, putting $u = |z_1|^2$ gives

$$\int_{S^3} \alpha_0 \wedge d\alpha_0 = 4\pi^2 a_1 a_2 \int_0^1 \frac{du}{[a_1^2 u + a_2^2(1-u)]^2} = \frac{4\pi^2}{a_1 a_2}.$$

The spheres are oriented outward. In equivariant normal coordinates, $g = \delta_4 + O(r^2)$ and $B = B_0 + O(r^2)$, where $r = d_g(p, \cdot)$. The curvature term in (15) has integral $O(r^2)$ on the sphere of radius r, and the remaining term converges to its Euclidean model. Thus

$$\lim_{\epsilon \downarrow 0} \int_{\partial B_g(p,\epsilon)} T_K = \frac{\mathrm{tr}(B_0^2)}{8\pi^2} \int_{S^3} \alpha_0 \wedge d\alpha_0 = -\left(\frac{a_1}{a_2} + \frac{a_2}{a_1}\right).$$

Since $\ell_K = O(r^{-1})$, the form T_K is locally integrable at p. Stokes's theorem, as in the Euler case, gives

$$dT_K = p_1(\nabla^g) - \left(\frac{a_1}{a_2} + \frac{a_2}{a_1}\right)\delta_p \quad \text{distributionally on } M. \tag{16}$$

Apply (16) to $K_s = -\omega^s X_1 + X_2$ and a cutoff χ_s equal to one on $\xi \le s$, zero on $\xi \ge 2s$, with $|d\chi_s|_g \le C/r_s$. The annular Killing estimate gives

$$\left|\int_M \chi_s p_1 + \omega^s + (\omega^s)^{-1}\right| \le C\left[r_s^{-2}\mathrm{Vol}_g(\mathcal{A}_s)^{1/2}\left(\int_{\mathcal{A}_s} |\mathrm{Rm}_g|_g^2 dV_g\right)^{1/2} + v_s\right] \to 0.$$

Here (14) and Proposition 3.9 justify the limit. Absolute integrability of p_1 now yields

$$-\omega^s - (\omega^s)^{-1} \to P.$$

For every nonzero real t, $|t + t^{-1}| \ge 2$. Thus $|P| \ge 2$, while $\mathcal{E}_+ + \mathcal{E}_- = 1$ gave $|P| \le 2$. Equality forces $W^+ = 0$ or $W^- = 0$.

After choosing orientation, the induced connection on Λ_g^+ is flat. Since M is simply connected, it has a global parallel orthonormal frame of self-dual forms. Scaling these forms to norm $\sqrt{2}$ gives a hyperkähler triple. The remaining classification is a consequence of the known complete hyperkähler circle classification: the isometric torus supplies the required tri-Hamiltonian circle as follows. The torus acts on this three-dimensional space through $SO(3)$. An abelian connected subgroup of $SO(3)$ has dimension at most one, so the kernel contains a closed circle preserving the triple. For its generator K, the one-forms $\iota_K \omega_i$ are closed by Cartan's formula and exact because $H^1(M; \mathbb{R}) = 0$. Thus this is a tri-Hamiltonian circle action.

The complete hyperkähler circle classification of Swann [19, Theorem 3.1], with the center description in Dancer–Swann [10, Section 2.2], applies to this complete tri-Hamiltonian circle action. In this description the empty center case has topology $S^1 \times \mathbb{R}^3$, and a nonempty center set Z has $b_2 = |Z| - 1$ (infinitely many centers give infinite second Betti number). Since M is homeomorphic to $\mathbb{R}^4$, there is exactly

one center. With the circle generator normalized to period 2π, the one-center Gibbons–Hawking potential is

$$V(x) = c_0 + \frac{1}{2|x - x_0|}, \qquad c_0 \geq 0,$$

where x is a moment-map coordinate on $\mathbb{R}^3$. The case $c_0 = 0$ is Euclidean and $c_0 > 0$ is Taub–NUT; all positive one-center parameters differ by constant scaling. The nonflat branch is therefore Taub–NUT. □

References

[1] S. Alexander, V. Kapovitch and A. Petrunin, *Invitation to Alexandrov geometry: CAT[0] spaces*.

[2] A. L. Besse, *Einstein Manifolds*, Ergebnisse der Mathematik und ihrer Grenzgebiete (3), vol. 10, Springer-Verlag, Berlin, 1987.

[3] M. R. Bridson and A. Haefliger, *Metric Spaces of Non-Positive Curvature*, Springer, 1999.

[4] J. Cheeger and G. Tian, *Curvature and injectivity radius estimates for Einstein 4-manifolds*, J. Amer. Math. Soc. **19** (2006), no. 2, 487–525.

[5] X. Chen and Y. Li, *On the geometry of asymptotically flat manifolds*, Geom. Topol. **25** (2021), no. 5, 2469–2572.

[6] S.-Y. Cheng, *Eigenvalue comparison theorems and its geometric applications*, Math. Z. **143** (1975), 289–297.

[7] S.-S. Chern, *A simple intrinsic proof of the Gauss–Bonnet formula for closed Riemannian manifolds*, Ann. of Math. (2) **45** (1944), no. 4, 747–752.

[8] S.-S. Chern, *On the curvatura integra in a Riemannian manifold*, Ann. of Math. (2) **46** (1945), no. 4, 674–684.

[9] T. H. Colding, *New monotonicity formulas for Ricci curvature and applications. I*, Acta Math. **209** (2012), 229–263.

[10] A. Dancer and A. Swann, *Hypertoric manifolds and hyperKähler moment maps*, in *Special Metrics and Group Actions in Geometry*, Springer INdAM Series, vol. 23, Springer, Cham, 2017, 107–127.

[11] G. W. Gibbons and S. W. Hawking, *Gravitational multi-instantons*, Phys. Lett. B **78** (1978), no. 4, 430–432.

[12] A. Hatcher, *The Kirby torus trick for surfaces*, arXiv:1312.3518v3.

[13] S. W. Hawking, *Gravitational instantons*, Phys. Lett. A **60** (1977), no. 2, 81–83.

[14] M. Li and S. Sun, *Compact Toric Einstein Four-Manifolds*, manuscript.

[15] M. Li and S. Sun, *Gravitational instantons and harmonic maps*, arXiv:2507.15284v1, 2025.

[16] P. Li and S.-T. Yau, *On the parabolic kernel of the Schrödinger operator*, Acta Math. **156** (1986), 153–201.

[17] T. Liu, *Toric Einstein 4-manifolds with non-negative sectional curvature*, Ann. Global Anal. Geom. **67** (2025), article 14.

[18] P. Orlik and F. Raymond, *Actions of the torus on 4-manifolds. I*, Trans. Amer. Math. Soc. **152** (1970), 531–559.

[19] A. Swann, *Twists versus modifications*, Adv. Math. **303** (2016), 611–637.

[20] B. Weber, *Energy and asymptotics of Ricci-flat 4-manifolds with a Killing field*, Proc. Amer. Math. Soc. **147** (2019), 3117–3130.

[21] S.-T. Yau, *Harmonic functions on complete Riemannian manifolds*, Comm. Pure Appl. Math. **28** (1975), 201–228.